

The School of Mathematics



THE UNIVERSITY
of EDINBURGH

Dyson Brownian Motion and Sampling from Invariant Matrix Ensembles

by

Fred Dickinson

Dissertation Presented for the Degree of
MSc in Computational Applied Mathematics

August 2026

Supervised by
Konstantinos Zygalakis

Abstract

Dyson Brownian motion is a multi-dimensional Brownian motion in which trajectories are conditioned to be non-intersecting. From the perspective of statistical mechanics, it describes the stochastic evolution of points on a line according to mutual logarithmic repulsion and a confinement potential. There is a surprising connection between random matrix theory and Dyson Brownian motion in that its unique stationary distribution is the eigenvalue distribution of a large class of random matrix models: the β -ensembles. Simulation of its dynamics can therefore act as a sampling algorithm for these ensembles, providing insight into their finite-dimensional spectral statistics and their convergence to a limiting law. In this dissertation we study a family of Langevin-based Markov chain Monte Carlo algorithms for this sampling problem through a combination of theoretical analysis and numerical experiments. We demonstrate that macroscopic eigenvalue statistics of the β -ensembles can be recovered through the implicit Langevin algorithm with quadratic complexity in the dimension of the ensemble. For finer statistics, such as local eigenvalue spacings or fluctuations at the edge of the spectrum, we propose the use of Metropolis-adjusted algorithms that can be of sub-cubic complexity when paired with fast multipole methods. More broadly, we establish the utility of Dyson Brownian motion as a versatile sampling scheme capable of targeting a wide variety of ensembles that are beyond the scope of alternative algorithms.

Acknowledgments

My thanks go to Kostas for continually going above and beyond as an excellent supervisor; to Theo for his time and insight into random matrices; to Leroy for his unfailing support; and, above all, to Christina for putting up with me.

Own Work Declaration

I confirm that all of this work is my own except where indicated.

Generative AI Disclosure

Generative AI tools (Gemini, Claude) have been used for the purposes of debugging and refactoring code for this project. All AI output has been reviewed and verified.

No AI tools have been used for the writing of this report.

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1 Introduction

A central object in the study of stochastic processes is Brownian motion, simply understood as the random movement of particles in an ambient space. More technically, a one-dimensional Brownian motion with initial position μ can be defined by the collection of random variables $\{W_t\}_{t \geq 0}$ satisfying the following four properties [87].

1. $W_0 = \mu$ almost surely.
2. The map $t \mapsto W_t$ is almost surely continuous.
3. For all $0 \leq s < t$, the increment $W_t - W_s$ has distribution $\mathcal{N}(0, t - s)$.
4. For all $t_0 \leq t_1 \leq \dots \leq t_N$, the increments $W_{t_j} - W_{t_{j-1}}$ are jointly independent for all $j = 1, 2, \dots, N$.

Extending this definition to higher dimensions is done by considering the vector of independent Brownian motions $W_t = (W_t^{(1)}, W_t^{(2)}, \dots, W_t^{(d)})$.

There are several close relatives of the standard Brownian motion. For example, fractional Brownian motion introduces ‘memory’ into the trajectories, and the Brownian bridge conditions the start and end points of the trajectories to be fixed and equal. Of most importance to us, however, is the Dyson Brownian motion (DBM). This extension considers an N -dimensional Brownian motion with the introduction of level repulsion so that no two trajectories cross paths. For an illustration, see Figure 1. Formally, let $\lambda = (\lambda_1, \dots, \lambda_N)$ lie inside the Weyl chamber $\mathbb{R}_>^N := \{\lambda \in \mathbb{R}^N : \lambda_1 < \dots < \lambda_N\}$. Then DBM is the solution to the coupled stochastic differential equations

$$d\lambda_k = \sum_{j \neq k} \frac{dt}{\lambda_k - \lambda_j} + \sqrt{\frac{2}{\beta}} dW_k, \quad k = 1, \dots, N, \quad (1.1)$$

where $W_1, \dots, W_N$ are independent Brownian motions and $\beta > 0$ is the Dyson index acting as an inverse temperature [87]. Typically we consider (1.1) with mean-field scaling and a confining potential V so that

$$d\lambda_k = \left(\frac{1}{N} \sum_{j \neq k} \frac{1}{\lambda_k - \lambda_j} - \frac{1}{2} V'(\lambda_k) \right) dt + \sqrt{\frac{2}{\beta N}} dW_k \quad (1.2a)$$

$$= -\partial_{\lambda_k} H_N(\lambda) + \sqrt{\frac{2}{\beta N}} dW_k \quad (1.2b)$$

for $k = 1, \dots, N$, where the system’s energy or Hamiltonian $H_N(\lambda)$ is given by

$$H_N(\lambda) = \frac{1}{2} \sum_{j=1}^N V(\lambda_j) - \frac{1}{N} \sum_{j < k} \log(\lambda_k - \lambda_j). \quad (1.3)$$

It is common to view (1.2) from a statistical mechanics perspective, wherein one considers the stochastic evolution of N particles $\lambda_1, \dots, \lambda_N$ on a line according to mutual logarithmic repulsion and a confining potential V .

There are several aspects of DBM that make it an interesting object to study. We focus on its connection to random matrix theory, the context in which it was first explored by Freeman Dyson [26]. As will be discussed in Chapter 2, the equilibrium measure

$$\frac{1}{Z_{N,\beta}} \exp(-\beta N H_N(\lambda)), \quad Z_{N,\beta} = \int_{\mathbb{R}^N} \exp(-\beta N H_N(\lambda)) d\lambda_1 d\lambda_2 \dots d\lambda_N \quad (1.4)$$

is the unique equilibrium measure for (1.2) but also describes the joint probability density function (jpdf) for the eigenvalues of certain classes of random matrices [33]. Therefore

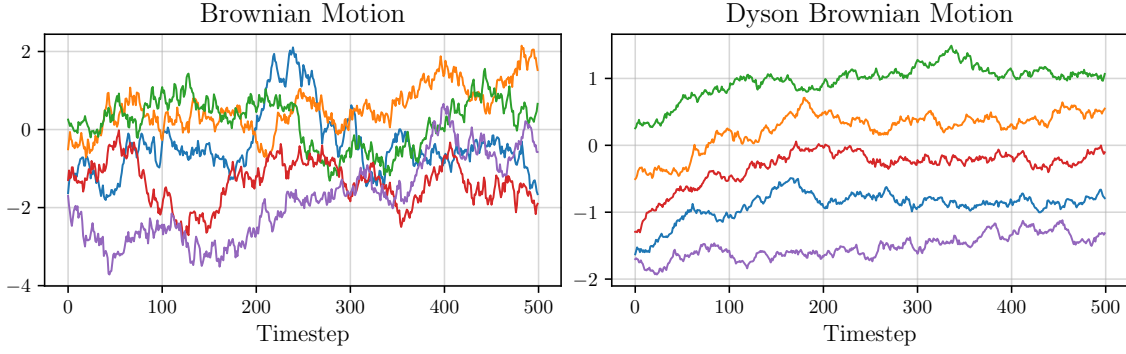


Figure 1: An illustration of $N = 5$ trajectories of Brownian motion (left) and Dyson Brownian motion (right) subject to a quadratic confining potential. The Dyson index is set to $\beta = 4$ to accentuate the level repulsion.

DBM can act as a sampling scheme for random matrices through the simulation of the Langevin dynamics (1.2). In fact, it is a uniquely desirable sampling scheme because it applies to a far wider range of random matrices than alternative algorithms [25, 69, 36].

The objective of this dissertation is thus twofold. First, we aim to provide an accessible exposition into random matrix theory. This is contained in Chapter 2 and is primarily based on the sources [87, 61, 33, 58, 30, 1]. Second, in Chapter 3 we investigate sampling schemes for random matrices based on the dynamics (1.2). Langevin-based algorithms are generally well studied, and it is the logarithmic repulsion interaction of DBM that produces computational difficulty - mainly owing to its highly singular nature, but also because of its inherent quadratic computational complexity. We include a review of two algorithms that have already been proposed in the literature [57, 18], analysis of the system, and discussion of the application of other algorithms that include more recent developments in the literature. A range of numerical experiments for these algorithms is then given in Chapter 4 before a concluding discussion in Chapter 5.

2 On Random Matrix Theory

Random matrix theory has had a multitude of recent developments that reach into many branches of mathematics and physics. In this chapter we introduce several key areas of the field, particularly those that have a connection to Dyson Brownian motion (1.2). Our presentation is intended to be introductory and accessible; for a more complete treatment, see, for instance, [87, 61, 33, 58, 30, 1].

2.1 The Gaussian and Beta Ensembles

At the most fundamental level, random matrices are matrices whose entries are random variables. Matrix ensembles are collections of random matrices that share a particular probability distribution. For example, a matrix $M \in \mathbb{R}^{N \times N}$ constructed with variance σ^2 so that

$$M_{kk} \sim \mathcal{N}(0, 2\sigma^2), \quad M_{k\ell} \sim \mathcal{N}(0, \sigma^2), \quad k < \ell, \quad M_{\ell k} = M_{k\ell}, \quad (2.1)$$

belongs to the Gaussian Orthogonal Ensemble (GOE). An equivalent construction takes a matrix X with independent entries all drawn from $\mathcal{N}(0, \sigma^2)$ and constructs $M = (X + X^T)/\sqrt{2}$. The main consequence of σ^2 is the scaling of the eigenvalues; they concentrate on the interval $[-2\sqrt{N}, 2\sqrt{N}]$ when $\sigma^2 = 1$ and on $[-2, 2]$ when $\sigma^2 = 1/N$ [33].

An important component of this construction is that the diagonal entries of M have twice the variance of the off-diagonal entries. To understand why, first consider the joint probability density function (jpdf) of all entries of M ,

$$\begin{aligned} \mathbb{P}(M) &\propto \prod_{k=1}^N \mathbb{P}(M_{kk}) \prod_{k < \ell} \mathbb{P}(M_{k\ell}) \propto \prod_{k=1}^N \exp\left(-\frac{M_{kk}^2}{4\sigma^2}\right) \prod_{k < \ell} \exp\left(-\frac{M_{k\ell}^2}{2\sigma^2}\right) \\ &= \exp\left(-\frac{1}{4\sigma^2} \left[\sum_{k=1}^N M_{kk}^2 + 2 \sum_{k < \ell} M_{k\ell}^2 \right]\right) \\ &= \exp\left(-\frac{1}{4\sigma^2} \text{tr}(M^2)\right), \end{aligned} \quad (2.2)$$

because when M is symmetric the trace $\text{tr}(\cdot)$ satisfies

$$\text{tr}(M^2) = \sum_{k=1}^N \sum_{\ell=1}^N M_{k\ell} M_{\ell k} = \sum_{k=1}^N \sum_{\ell=1}^N M_{k\ell}^2 = \sum_{k=1}^N M_{kk}^2 + 2 \sum_{k < \ell} M_{k\ell}^2.$$

It is well known that the trace is invariant under orthogonal transformation, e.g. by using the cyclic property $\text{tr}(ABC) = \text{tr}(BCA)$ to see $\text{tr}(OMO^T) = \text{tr}(MO^T O) = \text{tr}(M)$ for any orthogonal matrix O . Therefore the joint density (2.2) of M , when constructed with independent Gaussians as in (2.1), is invariant under orthogonal transformation in the sense that $\mathbb{P}(M) = \mathbb{P}(OMO^T)$.

The combination of both (a) invariance under orthogonal (or more generally, unitary) transformation and (b) independence of entries makes the Gaussian ensembles particularly special. In fact, they are the only ensembles that can simultaneously satisfy both of these criteria [58] and for this reason will often be the focus of this project. For definiteness, the three classical Gaussian ensembles are described below.

1. The Gaussian Orthogonal Ensemble (GOE) is the ensemble of real symmetric matrices whose entries are distributed

$$M_{kk} \sim \mathcal{N}(0, 2\sigma^2), \quad M_{k\ell} \sim \mathcal{N}(0, \sigma^2), \quad k < \ell, \quad M_{\ell k} = M_{k\ell}.$$

Matrices from the GOE satisfy $\mathbb{P}(M) = \mathbb{P}(OMO^T)$ for any orthogonal matrix O

(the joint density is invariant under orthogonal transformation).

2. The Gaussian Unitary Ensemble (GUE) is the ensemble of Hermitian matrices whose entries are complex numbers $a + bi$ distributed

$$M_{kk} \sim \mathcal{N}(0, 2\sigma^2), \quad M_{k\ell} \sim \mathcal{N}(0, \sigma^2/2) + i\mathcal{N}(0, \sigma^2/2), \quad k < \ell, \quad M_{\ell k} = \overline{M_{k\ell}}.$$

Matrices from the GUE satisfy $\mathbb{P}(M) = \mathbb{P}(UMU^*)$ for any unitary matrix U (the joint density is invariant under unitary transformation).

3. The Gaussian Symplectic Ensemble (GSE) is the ensemble of self-dual matrices whose entries are quaternions $a + \mathbf{b}\mathbf{i} + \mathbf{c}\mathbf{j} + \mathbf{d}\mathbf{k}$ distributed

$$M_{kk} \sim \mathcal{N}(0, 2\sigma^2), \\ M_{k\ell} \sim \mathcal{N}(0, \sigma^2/4) + \mathbf{i}\mathcal{N}(0, \sigma^2/4) + \mathbf{j}\mathcal{N}(0, \sigma^2/4) + \mathbf{k}\mathcal{N}(0, \sigma^2/4), \quad k < \ell,$$

again with $M_{\ell k} = \overline{M_{k\ell}}$, but now $\overline{(\cdot)}$ indicates the quaternionic conjugate. Matrices from the GSE satisfy $\mathbb{P}(M) = \mathbb{P}(SM S^\dagger)$ for any symplectic matrix S (the joint density is invariant under symplectic transformation).

We can isolate the eigenvalues $\lambda = (\lambda_1, \dots, \lambda_N)$ of matrices from the Gaussian ensembles by considering the change of variables $M = Q\Lambda Q^\dagger$ for the appropriate orthogonal, unitary, or symplectic matrix Q . Applying this to the joint density, e.g. (2.2) for the GOE and the analogous cases for the GUE and GSE, yields the eigenvalue jpdf

$$\rho(\lambda_1, \dots, \lambda_N) \propto \exp\left(-\frac{\beta}{4\sigma^2} \sum_{j=1}^N \lambda_j^2\right) \prod_{j < k} |\lambda_k - \lambda_j|^\beta. \quad (2.3)$$

See [33] for a complete derivation. Informally, the trace becomes the sum of squared eigenvalues and the Vandermonde determinant

$$\Delta_N(\lambda) := \prod_{j < k} (\lambda_k - \lambda_j) \quad (2.4)$$

is the Jacobian arising from the change of variables. The parameter β is the Dyson index with $\beta \in \{1, 2, 4\}$ corresponding to the GOE, GUE, and GSE respectively. From a structural point of view, the parameter β describes the number of independent degrees of freedom in each matrix entry. It is also precisely the same parameter appearing in Dyson Brownian motion (1.1). To see this, set $\sigma^2 = 1/N$ and use the identity

$$\prod_i y_i^\beta = \exp\left(\beta \sum_i \log y_i\right)$$

to achieve

$$\rho(\lambda_1, \dots, \lambda_N) \propto \exp\left(-\beta N \left(\frac{1}{2} \sum_{j=1}^N \frac{\lambda_j^2}{2} - \frac{1}{N} \sum_{j < k} \log |\lambda_k - \lambda_j|\right)\right), \quad (2.5)$$

another form of the probability distribution of eigenvalues from the Gaussian ensembles. This is the same form as the equilibrium measure (1.4) with a quadratic potential, i.e. $V(x) = x^2/2$. Indeed, if we replace $\sum_{j=1}^N \lambda_j^2/2$ with a functional form $\sum_{j=1}^N V(\lambda_j)$ then we obtain the density

$$\exp\left(-\beta N \left(\frac{1}{2} \sum_{j=1}^N V(\lambda_j) - \frac{1}{N} \sum_{j < k} \log |\lambda_k - \lambda_j|\right)\right) = \exp(-\beta N H_N(\lambda)), \quad (2.6)$$

where $H_N(\lambda)$ is the Hamiltonian as in (1.3). Up to the normalising constant, (2.6) is the probability distribution for the central ‘ β -ensembles’. It is clear that these encompass the distribution of eigenvalues from the Gaussian ensembles when $\beta \in \{1, 2, 4\}$ and the potential V is quadratic, $V(x) = x^2/2$. For a general $\beta > 0$ and quadratic potential the ensembles are known as the β -Hermite ensembles.

It is important to be precise here. In full generality, one can consider both real and complex λ_j . That is, $\lambda_j \in \mathbb{R}^d$ for $d \in \{1, 2\}$, where we view $\mathbb{C} \cong \mathbb{R}^2$. The β -ensembles then represent a point process on $\mathbb{R}^d$ for all $\beta > 0$ rather than a model of random matrices themselves, and only for $\beta \in \{1, 2, 4\}$ are they realised as the eigenvalue distribution of classical random matrices. For $d = 1$ a relevant assertion is ‘Dyson’s threefold way’ [27]: that random matrices with independent entries that are also invariant under conjugation must have $\beta \in \{1, 2, 4\}$. In fact, as previously revealed, it is only the Gaussian ensembles that can simultaneously satisfy these criteria [58]. If one drops the invariance requirement it is possible to see the β -ensembles as the eigenvalues of random tridiagonal matrices [25], albeit without full generality of the potential V . We discuss these models and the general problem of sampling from (2.6) in Chapter 3.

2.2 A Coulomb Gas Perspective

One natural interpretation for the density (2.6) comes from statistical mechanics, where the eigenvalues $\lambda = (\lambda_1, \dots, \lambda_N)$ are considered as N points in $\mathbb{R}^d$ configured according to (a) mutual logarithmic repulsion and (b) a confining potential V . Here $d \geq 1$ is the dimension of each eigenvalue and again we view $\mathbb{C} \cong \mathbb{R}^2$. In this setting, β acts as an inverse temperature, H_N is the energy of the system, and

$$\frac{1}{Z_{N,\beta}} \exp(-\beta N H_N(\lambda)) d\lambda$$

is seen as a Gibbs measure with the normalising factor

$$Z_{N,\beta} = \int_{(\mathbb{R}^d)^N} \exp(-\beta N H_N(\lambda)) d\lambda_1 d\lambda_2 \cdots d\lambda_N. \quad (2.7)$$

For general potentials V it is difficult to compute (2.7). A particularly well studied case is when V is quadratic and $d = 1$, in which $Z_{N,\beta}$ can be found through the Selberg integral; see, for instance, [33, Chapter 4].

This perspective is just one example from a more general framework. We follow the notation of Serfaty [83] who considers N points $(x_1, \dots, x_N)$, each in $\mathbb{R}^d$, distributed according to the energy functionals $\mathcal{H}_N : (\mathbb{R}^d)^N \rightarrow (-\infty, \infty]$ of the form

$$\mathcal{H}_N(x) = N \sum_{j=1}^N V(x_j) + \frac{1}{2} \sum_{j \neq k} g(x_k - x_j), \quad (2.8)$$

where $V : \mathbb{R}^d \rightarrow (-\infty, \infty)$ is called the ‘external field’ and $g : \mathbb{R}^d \rightarrow (-\infty, \infty]$ the ‘pair interaction potential’. Typically g is defined

$$g(x) = \begin{cases} \frac{1}{s} |x|^{-s}, & s \neq 0, \\ -\log |x|, & s = 0, \end{cases} \quad (2.9)$$

with $|x|$ indicating the Euclidean norm on $\mathbb{R}^d$, reducing to the absolute value when $d = 1$. The first case corresponds to particle systems known as Riesz gases, and the second log-

gases. The relationship between the two can be understood through the limit

$$-\log |x| = \lim_{s \rightarrow 0} \frac{1}{s} (|x|^{-s} - 1).$$

All cases of the interaction (2.9) produce a repulsive force between $x = (x_1, \dots, x_N)$ and motivate the need for the confinement potential V . Of particular interest to us is the Coulomb case $s = d - 2$, for which the system is known as a Coulomb gas. This corresponds to the logarithmic case when $d = 2$, and indeed we view the repulsion in the Hamiltonian for our system (1.3),

$$H_N(\lambda) = \frac{1}{2} \sum_{j=1}^N V(\lambda_j) - \frac{1}{N} \sum_{j < k} \log |\lambda_k - \lambda_j|,$$

as a two-dimensional Coulomb kernel despite the individual particles themselves possibly being restrained to $\mathbb{R}$. Here we will not expand on the interactions between Riesz, Coulomb, and log-gases and recommend [18, 83, 33] for a comprehensive discussion.

Let us briefly address some discrepancies between our Hamiltonian and the general framework (2.8). The Gibbs measure for the energy functionals $\mathcal{H}_N$, in our case of $s = 0$, is proportional to $\exp(-\beta \mathcal{H}_N)$. Direct comparison to the Gibbs measure $\exp(-\beta N H_N(\lambda))$ makes it clear that there is no factor of N missing. Moreover, the choice of one half before the confinement potential is merely a consequence of the scaling of V , and the symmetric kernel $\log |\lambda_k - \lambda_j|$ allows us to write the repulsion over indices $j < k$.

Understanding the density (2.6) as a Coulomb gas is an excellent tool for its analysis. For example, the empirical density of the eigenvalues

$$\mu_N := \frac{1}{N} \sum_{j=1}^N \delta_{\lambda_j} \tag{2.10}$$

is a common object of study. When V is quadratic, $V(x) = x^2/2$, this is known to converge in the limit $N \rightarrow \infty$ to Wigner's semicircle law

$$\lim_{N \rightarrow \infty} \mu_N(x) = \frac{1}{2\pi} \sqrt{4 - x^2} \mathbb{I}(|x| \leq 2) =: \rho_{\text{SC}}(x), \tag{2.11}$$

and this can be shown analytically through the Coulomb gas; see e.g. [58, Chapter 4] for an accessible proof. There are several other methods of proof [87, 61, 33, 30], a few of which are outlined later in this text. To illustrate this law we numerically calculate the eigenvalues of a matrix from the GUE (refer to Section 2.1),

$$M_{kk} \sim \mathcal{N}\left(0, \frac{2}{N}\right), \quad M_{k\ell} \sim \mathcal{N}\left(0, \frac{1}{2N}\right) + i\mathcal{N}\left(0, \frac{1}{2N}\right), \quad k < \ell, \quad M_{\ell k} = \overline{M_{k\ell}},$$

with $N = 5000$. A histogram of its eigenvalues can be seen in Figure 2.

The limiting distribution of the empirical density (2.10) is often described as the ‘macroscopic’ eigenvalue behaviour. Typical areas of study in random matrix theory also include the distribution of spacings between the eigenvalues (the ‘microscopic’ behaviour) and the distribution of the maximal eigenvalue - or, more generally, fluctuations on the edge of the spectrum. We discuss these points in Sections 2.3, 2.4, and 2.5. An important aspect of all these points is that of universality. Simply put, this asserts that the limiting macroscopic behaviour of (2.6) depends only on V and is independent of β , whereas the microscopic behaviour depends only on β and is independent of V . This notion will appear several times in this project. See [29] for universality in random matrices and [20, 73] for the larger topic of ‘Kardar-Parisi-Zhang’ (KPZ) universality.

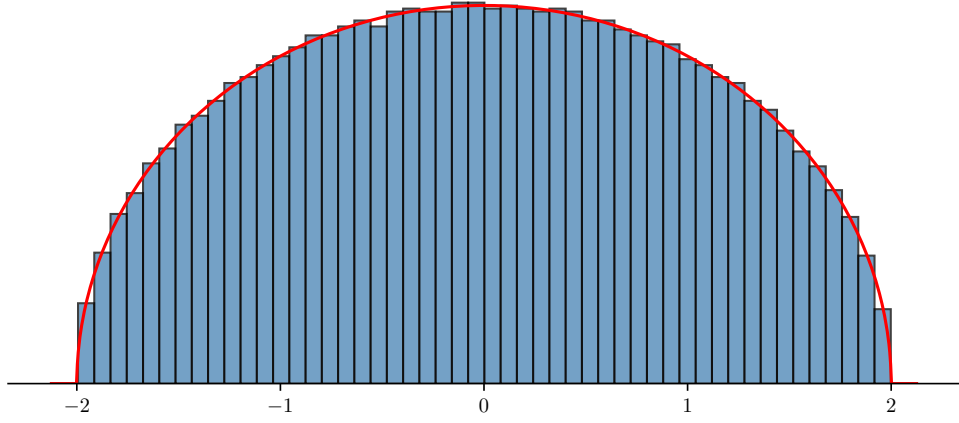


Figure 2: A histogram of the eigenvalues of a 5000×5000 random matrix from the Gaussian Unitary Ensemble. The red curve is the semicircle distribution.

2.3 Determinantal Structure

The eigenvalues $\lambda = (\lambda_1, \dots, \lambda_N)$ from the β -ensembles (2.6) can be understood as defining a point process [21] on $\mathbb{R}$ for every $\beta > 0$; that is, sets of random points on $\mathbb{R}$ governed by their k -point correlation functions $\rho_k(\lambda_1, \dots, \lambda_k)$. Loosely speaking, the correlation functions describe the probability of finding k points at chosen positions in a system. It turns out that $\beta \in \{1, 2, 4\}$ are once more special because, in these cases, the correlation functions ρ_k admit an exact closed form [33]: a Pfaffian for $\beta \in \{1, 4\}$ and a determinant when $\beta = 2$. These are known as Pfaffian and determinantal point processes respectively. In this section we follow the analysis presented in [87] to derive the determinantal structure for the β -ensembles when $\beta = 2$, and use these results to find the expected empirical spectral distribution for any finite N and potential V .

To begin, let $\lambda = (\lambda_1, \dots, \lambda_N)$ with $\lambda_j \in \mathbb{R}$ for all $j = 1, \dots, N$. Our starting point is the distribution of eigenvalues from the Gaussian ensembles (2.3) with $\beta = 2$ and $\sigma^2 = 1$ so that

$$\rho_N(\lambda) \propto \exp \left(- \sum_{j=1}^N \frac{\lambda_j^2}{2} \right) |\Delta_N(\lambda)|^2, \quad (2.12)$$

where we have swapped to the notation ρ_N to emphasise dependence on N , and $\Delta_N(\lambda)$ is the Vandermonde determinant (2.4). It perhaps shouldn't be surprising to learn that this is the main object needed to achieve the *determinantal* form of (2.12). Its name comes from the fact that it can be expressed as the determinant

$$\Delta_N(\lambda) = \det \begin{bmatrix} 1 & \lambda_1 & \lambda_1^2 & \cdots & \lambda_1^{N-1} \\ 1 & \lambda_2 & \lambda_2^2 & \cdots & \lambda_2^{N-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \lambda_N & \lambda_N^2 & \cdots & \lambda_N^{N-1} \end{bmatrix} =: \det(\mathcal{V}(\lambda)). \quad (2.13)$$

We denote the Vandermonde matrix as $\mathcal{V}$ to avoid confusion with the potential V . There are several ways to show (2.13). One approach uses a result from linear algebra: that adding a scalar multiple of one column to another does not affect the determinant. Indeed, let C_k denote the columns of $\mathcal{V}$ and transform $C_k \mapsto C_k - \lambda_1 C_{k-1}$ for $k = N, N-1, \dots, 2$

so that

$$\Delta_N(\lambda) = \det \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & \lambda_2 - \lambda_1 & \lambda_2(\lambda_2 - \lambda_1) & \cdots & \lambda_2^{N-2}(\lambda_2 - \lambda_1) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \lambda_N - \lambda_1 & \lambda_N(\lambda_N - \lambda_1) & \cdots & \lambda_N^{N-2}(\lambda_N - \lambda_1) \end{bmatrix}.$$

Then apply cofactor expansion along the top row and use the multilinearity of the determinant to factorise the common factor in each row,

$$\begin{aligned} \Delta_N(\lambda) &= \det \begin{bmatrix} \lambda_2 - \lambda_1 & \lambda_2(\lambda_2 - \lambda_1) & \cdots & \lambda_2^{N-2}(\lambda_2 - \lambda_1) \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_N - \lambda_1 & \lambda_N(\lambda_N - \lambda_1) & \cdots & \lambda_N^{N-2}(\lambda_N - \lambda_1) \end{bmatrix} \\ &= (\lambda_2 - \lambda_1)(\lambda_3 - \lambda_1) \cdots (\lambda_N - \lambda_1) \det \begin{bmatrix} 1 & \lambda_2 & \cdots & \lambda_2^{N-2} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \lambda_N & \cdots & \lambda_N^{N-2} \end{bmatrix}, \end{aligned}$$

and observe how iteratively repeating this process achieves the product form (2.4).

In general, one can transform the entries of $\mathcal{V}$ to be any monic polynomial of choice by following the same argument. It turns out that the (probabilistic) Hermite polynomials $He_k(x)$ are particularly useful. These are monic polynomials of degree k , with $He_0(x) = 1$, $He_1(x) = x$, $He_2(x) = x^2 - 1$, for example, and satisfy the recurrence relation

$$He_{k+1}(x) = xHe_k(x) - kHe_{k-1}(x). \quad (2.14)$$

See [86] for a detailed discussion. Crucially, they are orthogonal with respect to $e^{-x^2/2}$. So, transforming $\mathcal{V}$ to

$$W := \begin{bmatrix} He_0(\lambda_1) & He_1(\lambda_1) & \cdots & He_{N-1}(\lambda_1) \\ He_0(\lambda_2) & He_1(\lambda_2) & \cdots & He_{N-1}(\lambda_2) \\ \vdots & \vdots & \ddots & \vdots \\ He_0(\lambda_N) & He_1(\lambda_N) & \cdots & He_{N-1}(\lambda_N) \end{bmatrix}$$

through only column operations, we have the chain of equalities

$$|\Delta_N(\lambda)|^2 = |\det(\mathcal{V})|^2 = \det(\mathcal{V}\mathcal{V}^T) = \det(WW^T).$$

By then considering $W = [He_j(\lambda_i)]_{1 \leq i, j \leq N}$, or by explicitly writing out the product WW^T , it follows that

$$|\Delta_N(\lambda)|^2 = \det \left[\sum_{k=0}^{N-1} He_k(\lambda_i) He_k(\lambda_j) \right]_{1 \leq i, j \leq N},$$

and thus all that is needed is to decompose

$$\exp \left(- \sum_{j=1}^N \frac{\lambda_j^2}{2} \right) = \prod_{i=1}^N \exp \left(- \frac{\lambda_i^2}{4} \right) \prod_{j=1}^N \exp \left(- \frac{\lambda_j^2}{4} \right)$$

and use the multilinearity of the determinant so that (2.12) becomes

$$\rho_N(\lambda) \propto \det \left[\sum_{k=0}^{N-1} He_k(\lambda_i) \exp \left(- \frac{\lambda_i^2}{4} \right) He_k(\lambda_j) \exp \left(- \frac{\lambda_j^2}{4} \right) \right]_{1 \leq i, j \leq N}.$$

Further analysis of this object can be greatly simplified by transitioning to the orthonormal

wavefunctions

$$\psi_k(x) = \frac{He_k(x)e^{-x^2/4}}{(2\pi)^{1/4}\sqrt{k!}},$$

because now $\rho_N(\lambda)$ can be written in terms of the kernel K_N ,

$$K_N(x, y) := \sum_{k=0}^{N-1} \psi_k(x)\psi_k(y), \quad (2.15)$$

a central object in random matrix theory. Indeed, the proportionality constant greatly simplifies (refer to [87, Chapter 2.6.4]) so that the density becomes

$$\rho_N(\lambda) = \frac{1}{N!} \det [K_N(\lambda_i, \lambda_j)]_{1 \leq i, j \leq N}. \quad (2.16)$$

There are several key properties of K_N . First, a consequence of the orthonormalisation is the trace identity

$$\int_{\mathbb{R}} K_N(x, x) dx = \int_{\mathbb{R}} \sum_{k=0}^{N-1} \psi_k(x)\psi_k(x) dx = \sum_{k=0}^{N-1} \int_{\mathbb{R}} \psi_k(x)^2 dx = N.$$

Moreover, it satisfies the reproducing formula

$$\begin{aligned} \int_{\mathbb{R}} K_N(x, z) K_N(z, y) dz &= \int_{\mathbb{R}} \sum_{j=0}^{N-1} \psi_j(x)\psi_j(z) \sum_{k=0}^{N-1} \psi_k(z)\psi_k(y) dz \\ &= \sum_{j=0}^{N-1} \sum_{k=0}^{N-1} \psi_j(x)\psi_k(y) \left(\int_{\mathbb{R}} \psi_j(z)\psi_k(z) dz \right) \\ &= \sum_{j=0}^{N-1} \sum_{k=0}^{N-1} \psi_j(x)\psi_k(y) \delta_{kj} = \sum_{k=0}^{N-1} \psi_k(x)\psi_k(y) = K_N(x, y), \end{aligned}$$

where δ_{kj} is the Kronecker delta. Together, these yield the determinantal integration formula (the Dyson-Gaudin integration lemma)

$$\int_{\mathbb{R}} \det [K_N(\lambda_i, \lambda_j)]_{1 \leq i, j \leq k+1} d\lambda_{k+1} = (N - k) \det [K_N(\lambda_i, \lambda_j)]_{1 \leq i, j \leq k}, \quad (2.17)$$

the proof of which can again be found in [87, Chapter 2.6.4]. Now, (2.17) is invaluable because it lets one realise the k -point correlation functions of the eigenvalues [84]. Recall that these describe the probability to find any k points at $\lambda_1, \dots, \lambda_k$; or, more technically, the probability density that k eigenvalues lie in infinitesimal intervals about $\lambda_1, \dots, \lambda_k$. In this case ρ_k can be found by considering the marginal distribution

$$\rho_k(\lambda_1, \dots, \lambda_k) := \frac{N!}{(N - k)!} \int_{\mathbb{R}} \cdots \int_{\mathbb{R}} \rho_N(\lambda_1, \dots, \lambda_N) d\lambda_{k+1} \cdots d\lambda_N \quad (2.18a)$$

$$= \frac{N!}{(N - k)!} \int_{\mathbb{R}} \cdots \int_{\mathbb{R}} \frac{1}{N!} \det [K_N(\lambda_i, \lambda_j)]_{1 \leq i, j \leq N} \quad (2.18b)$$

$$= \det [K_N(\lambda_i, \lambda_j)]_{1 \leq i, j \leq k}. \quad (2.18c)$$

The prefactor in (2.18a) is combinatorial, with the determinantal integration formula iteratively applied to achieve (2.18c).

Of particular interest to us is when $k = 1$, for which we obtain the expected empirical

density of the eigenvalues [87]

$$\mathbb{E}\mu_N = \frac{1}{N}K_N(x, x)dx \quad (2.19)$$

where μ_N is as defined in (2.10). So, to understand the eigenvalue density we just need to understand the kernel K_N . We continue our analysis by considering the recurrence relation for the orthonormal functions ψ_k [86],

$$x\psi_k(x) = \sqrt{k+1}\psi_{k+1}(x) + \sqrt{k}\psi_{k-1}(x). \quad (2.20)$$

We substitute this relation into the difference

$$x\psi_k(x)\psi_k(y) - y\psi_k(y)\psi_k(x),$$

taking the convention that $\psi_{-k} \equiv 0$ for all $k > 0$ and writing $a_k = \sqrt{k}$ for convenience, yielding

$$\begin{aligned} (x-y)\psi_k(x)\psi_k(y) &= a_{k+1}(\psi_{k+1}(x)\psi_k(y) - \psi_k(x)\psi_{k+1}(y)) \\ &\quad - a_k(\psi_k(x)\psi_{k-1}(y) - \psi_{k-1}(x)\psi_k(y)) \\ &=: a_{k+1}F_{k+1}(x, y) - a_kF_k(x, y). \end{aligned}$$

Notice the telescoping form. To make it clear,

$$(x-y)K_N(x, y) = \sum_{k=0}^{N-1} (a_{k+1}F_{k+1}(x, y) - a_kF_k(x, y)) = a_NF_N - a_0F_0,$$

and, since $a_0 = \sqrt{0} = 0$, we have

$$K_N(x, y) = \sqrt{N} \frac{\psi_N(x)\psi_{N-1}(y) - \psi_N(y)\psi_{N-1}(x)}{x-y}. \quad (2.21)$$

The identity (2.21) is more commonly seen as a particular application of the Christoffel-Darboux formula which holds for generic sequences of orthogonal polynomials [61]. For the expected empirical eigenvalue density (2.19) one can apply L'Hôpital's rule to achieve

$$K_N(x, x) = \sqrt{N}(\psi'_N(x)\psi_{N-1}(x) - \psi_N(x)\psi'_{N-1}(x)),$$

where ψ' indicates the derivative. It is possible to continue the analysis of K_N to obtain Wigner's semicircle law: there are several approaches detailed in [87], or one can use the Plancherel-Rotach asymptotics for orthogonal polynomials [35].

We leave those details and instead comment on some important points not yet mentioned. First, there is no requirement that the potential is quadratic. Provided V grows at least logarithmically at infinity, there is a family of polynomials $\pi_k(x)$ orthogonal with respect to $e^{-V(x)}$ (see [86, Chapter 2.2]). These can, for instance, be found through the Gram-Schmidt process or the Stieltjes recurrence relation [37]. Moreover, while it is true that the limiting macroscopic eigenvalue behaviour is independent of β , the same cannot be said for when N is finite. Thus the expected empirical density (2.19) is only exact for $\beta = 2$, and for this case we illustrate some examples in Figure 3. For $\beta \in \{1, 4\}$ one may instead use the Pfaffian structure of the correlation functions [33]. However, we note that at least for the classical potentials, the finite density converges to the limiting distribution very quickly; we return to this point and further details of the densities in Section 4.1.

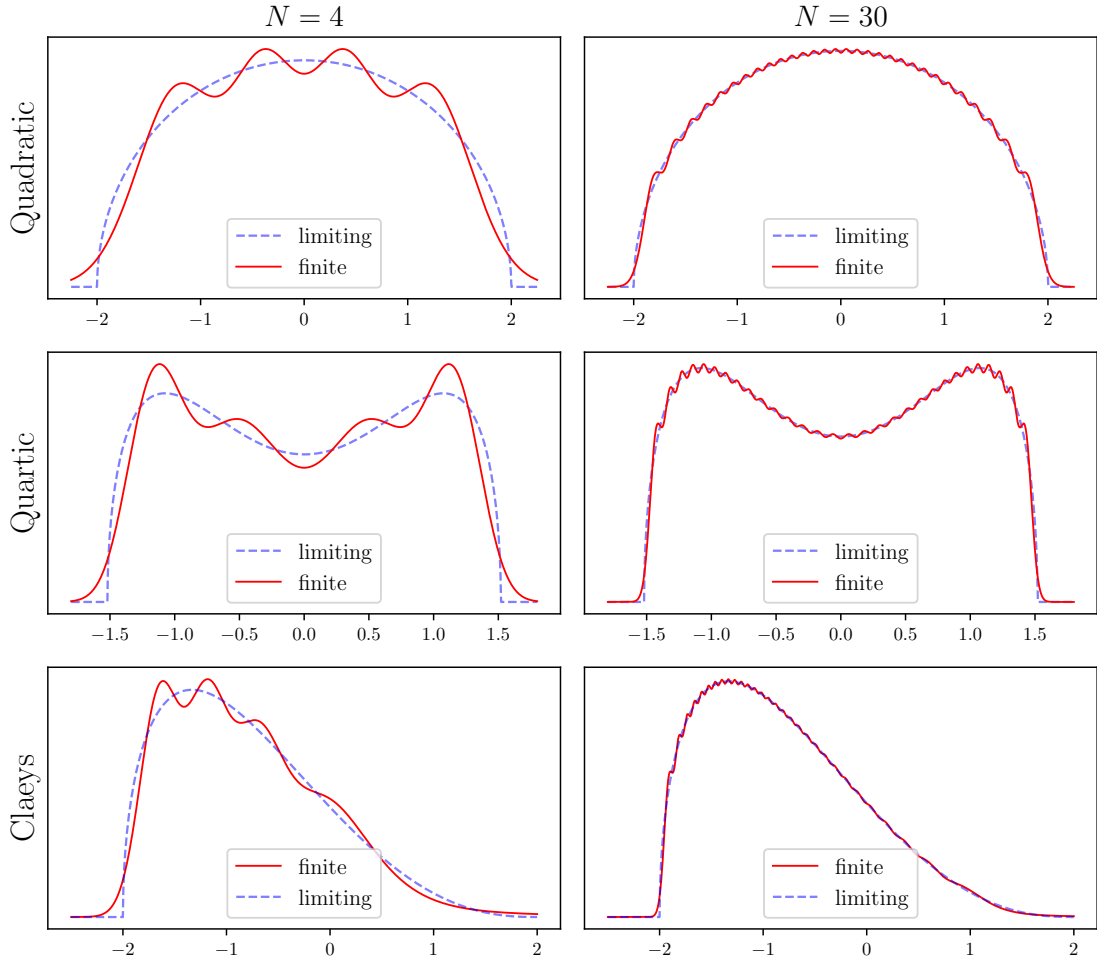


Figure 3: The finite-dimensional expected empirical density of the β -ensembles for the quadratic case, $V(x) = x^2/2$, the quartic case, $V(x) = x^4/4$, and the ‘Claeys’ case $V(x) = x^4/20 - 4x^3/15 + x^2/5 + 8x/5$, named for the lead author [19]. In all cases $\beta = 2$ and the density is obtained through numerical evaluation of the kernel K_N .

2.4 Microscopic Behaviour and Wigner’s Surmise

Now we briefly turn our attention to the eigenvalue behaviour on the microscopic level - in particular, the local spacings between them. These are interesting statistics to study because of the mutual eigenvalue repulsion, as seen from the jpdf (2.3) through the product

$$\prod_{j < k} |\lambda_k - \lambda_j|^\beta.$$

When β is small, e.g. $\beta = 1$, the repulsion is weaker and the particles will, on average, be much closer to each other than when the repulsion is strong, e.g. $\beta = 4$. Recall that $\beta \in \{1, 2, 4\}$ are of primary interest because they correspond to the orthogonal, unitary, and symplectic ensembles, but also because their k -point correlation functions admit an exact closed form - a Pfaffian for $\beta \in \{1, 4\}$ and a determinant for $\beta = 2$.

Of course, when $N \rightarrow \infty$ the average spacing between two eigenvalues becomes increasingly small, and it thus is important to scale these gaps accordingly. This is often referred to as the ‘unfolding’ of the spectrum. If $\lambda_1 < \lambda_2 < \dots < \lambda_N$, one way to do so is to map

$$\xi_j = NF(\lambda_j), \quad j = 1, \dots, N, \quad (2.22)$$

where F is the cumulative distribution function (cdf) for the eigenvalue jpdf. When the potential is quadratic, the limiting density is the semicircle distribution (2.11) whose cdf has the closed form [58]

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \left(\frac{x\sqrt{4-x^2}}{4} + \arcsin\left(\frac{x}{2}\right) \right),$$

and more generally one can use the kernel (2.15) to obtain the cdf for any given potential,

$$F_N(x) = \frac{1}{N} \int_{-\infty}^x K_N(s, s) ds.$$

Normalising the eigenvalues as in (2.22) ensures the mean gap $\xi_{j+1} - \xi_j$ is one [61, Chapter 5]. For the purpose of our discussion we construct three sets of 100×100 matrices from the GOE, GUE, and GSE (see the description in Section 2.1) numerically, calculate their eigenvalues, apply the scaling $\xi_j = NF(\lambda_j)$, and plot a histogram of the spacings between consecutive pairs. This is shown in Figure 4. One can observe the repulsion strength increasing as β increases. There also appears to be some distribution that the eigenvalues follow, which is precisely the content of Wigner's surmise: an approximation to the spacing distribution taking the form [43]

$$\delta_\beta(s) = a_\beta s^\beta \exp(-b_\beta s^2), \quad (2.23)$$

where $\delta_\beta(s)$ denotes the probability density of the gap $s := \xi_{j+1} - \xi_j$ for any pair of consecutive eigenvalues, and

$$a_\beta = 2 \frac{\Gamma^{\beta+1}((\beta+2)/2)}{\Gamma^{\beta+2}((\beta+1)/2)}, \quad b_\beta = \frac{\Gamma^2((\beta+2)/2)}{\Gamma^2((\beta+1)/2)}.$$

For $\beta \in \{1, 2, 4\}$ it is convenient to note

$$a_1 = \frac{\pi}{2}, \quad b_1 = \frac{\pi}{4}, \quad a_2 = \frac{32}{\pi^2}, \quad b_2 = \frac{4}{\pi}, \quad a_4 = \frac{262144}{729\pi^3}, \quad b_4 = \frac{64}{9\pi}.$$

It is reiterated that this form (2.23) is only a (rather good) approximation to the spacing distribution, obtained from performing calculations on 2×2 matrices. An exact formulation can be found by evaluating a Fredholm determinant involving the kernel K_N ; see [22, Section 5.4] for analysis and e.g. [11] for efficient numerical implementation.

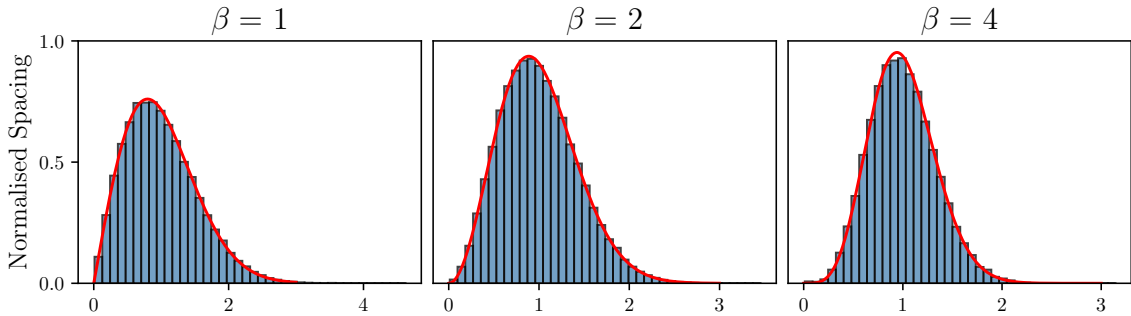


Figure 4: Spacings between the eigenvalues of 100×100 matrices from the GOE ($\beta = 1$), GUE ($\beta = 2$), and GSE ($\beta = 4$) after being normalised to have mean one. Wigner's surmise for each case is superimposed in red.

2.5 Edge Fluctuation and the Tracy-Widom Distribution

Thus far we have discussed the eigenvalue jpdf for the β -ensembles (2.6) from the macroscopic and microscopic points of view. A third and final perspective we take in this project concerns the behaviour on the edge of the spectrum. Theory on this topic can be quite complex and correspondingly we only provide insight into the key ideas at play, primarily based on the original work [32, 89, 90] and the books [33, 1].

As usual, we focus on the Gaussian ensembles normalised so that the limiting distribution of the eigenvalues has support $[-2, 2]$. The topic of edge fluctuation is concerned with the behaviour of the extreme eigenvalues as $N \rightarrow \infty$. To start, it is known that the distance between the largest eigenvalue and the spectrum edge is of order $N^{-2/3}$ (e.g. [1, Chapter 3]), so we define

$$x = N^{2/3}(\lambda_{\max} - 2) \quad (2.24)$$

as a centred and scaled random variable for the maximal eigenvalue. The cumulative distribution function for x in the limit $N \rightarrow \infty$ is precisely the Tracy-Widom distribution,

$$F_\beta(x) := \lim_{N \rightarrow \infty} \mathbb{P}(N^{2/3}(\lambda_{\max} - 2) \leq x). \quad (2.25)$$

Notice that (2.25) depends on β . For $\beta \in \{1, 2, 4\}$, a closed form for the law is given by Tracy and Widom [89, 90]. Let us comment on the case $\beta = 2$. An important piece here is the Airy kernel

$$K_{\text{Ai}}(x, y) = \frac{\text{Ai}(x)\text{Ai}'(y) - \text{Ai}'(x)\text{Ai}(y)}{x - y} \quad (2.26)$$

comprising the Airy function Ai , defined for real x as

$$\text{Ai}(x) = \frac{1}{\pi} \int_0^\infty \cos\left(\frac{t^3}{3} + xt\right) dt.$$

This case has been highlighted because the Airy kernel (2.26) has a strong connection to the kernel (2.21). Indeed, recall the k -point correlation function

$$\rho_k(\lambda_1, \dots, \lambda_k) = \det [K_N(\lambda_i, \lambda_j)]_{1 \leq i, j \leq k}$$

and the kernel

$$K_N(x, y) = \sqrt{N} \frac{\psi_N(x)\psi_{N-1}(y) - \psi_N(y)\psi_{N-1}(x)}{x - y}$$

as discussed in Section 2.3. Recall once more that ρ_k can be seen as the probability density that k eigenvalues lie in infinitesimal intervals about $\lambda_1, \dots, \lambda_k$. Tracy and Widom show that one can rescale the kernel K_N about the spectrum edge (2.24) so that

$$\lim_{N \rightarrow \infty} \frac{1}{N^{1/6}} K_N(2 + xN^{-2/3}, 2 + yN^{-2/3}) = K_{\text{Ai}}(x, y).$$

This is helpful because the probability $\mathbb{P}(N^{2/3}(\lambda_{\max} - 2) \leq x)$ is precisely the probability of finding no eigenvalues in the interval $(2 + xN^{-2/3}, \infty)$. With some work to address subtle points of uniform convergence of the kernels [32], the cumulative distribution for the largest eigenvalue can be expressed in terms of the Fredholm determinant

$$F_2(x) = \det(I - K_{\text{Ai}})_{(x, \infty)} \quad (2.27)$$

$$= \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \int_x^\infty \cdots \int_x^\infty \det [K_{\text{Ai}}(s_i, s_j)]_{1 \leq i, j \leq k} ds_1 \cdots ds_k, \quad (2.28)$$

where we take the convention that the $k = 0$ term is one.

Another major contribution from Tracy and Widom was proving that the Fredholm determinant (2.27) has the closed form

$$F_2(x) = \exp \left(- \int_x^\infty (t-x)q(t)^2 dt \right),$$

where $q(t)$ is the solution to the differential equation $q'' = tq + 2q^3$ (a second Painlevé transcendent) with boundary $q(t) \sim \text{Ai}(t)$ at infinity [89]. Two years later it was shown that the related cases $\beta = 1, 4$ can be written in terms of F_2 [90],

$$F_1(x)^2 = F_2(x) \exp \left(- \int_x^\infty q(t) dt \right),$$

$$F_4(x/\sqrt{2})^2 = F_2(x) \left[\frac{1}{2} \exp \left(\frac{1}{2} \int_x^\infty q(t) dt \right) + \frac{1}{2} \exp \left(-\frac{1}{2} \int_x^\infty q(t) dt \right) \right]^2.$$

An illustration of the Tracy-Widom distribution for the three cases $\beta \in \{1, 2, 4\}$ is given in Figure 5. These curves have been obtained through the TRACYWIDOM Python package¹ which uses known interpolation tables [6] and asymptotics [12].

Our discussion has so far concerned only the quadratic potential $V(x) = x^2/2$. However, as for a large number of results in random matrix theory, edge fluctuation exhibits universality: the Tracy-Widom distributions F_β govern the edge fluctuations for a broad class of potentials satisfying modest assumptions. This was first shown for $\beta \in \{1, 2, 4\}$ by Deift et al. [23], and only much later was universality for any $\beta > 0$ proven [14]. The limiting distribution for generic $\beta > 0$ can alternatively be characterised as the smallest eigenvalue of the so-called ‘stochastic Airy operator’ [74]. Universality is therefore not just confined to the bulk of the spectrum as discussed in Section 2.2 and Section 2.4, but extends to its extremes as well.

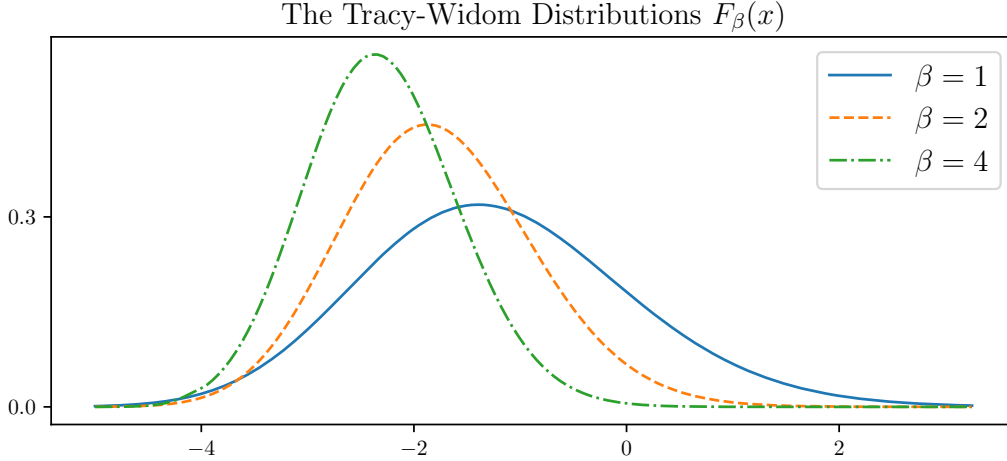


Figure 5: The Tracy-Widom distributions for the three cases $\beta \in \{1, 2, 4\}$. These curves have been obtained through the TRACYWIDOM Python package.

¹<https://github.com/yymao/TracyWidom>

2.6 Other Notable Ensembles

There are several notable matrix ensembles in random matrix theory, and it would not do the subject justice to focus solely on the β -ensembles. Recall that these assert ordering on the eigenvalues, $\lambda \in \mathbb{R}_{>}^N$, and are governed by the jpdf (2.6)

$$\rho(\lambda_1, \dots, \lambda_N) \propto \exp(-\beta N H_N(\lambda))$$

where the Hamiltonian is given by

$$H_N(\lambda) = \frac{1}{2} \sum_{j=1}^N V(\lambda_j) - \frac{1}{N} \sum_{j < k} \log(\lambda_k - \lambda_j).$$

The β -ensembles encompass the Gaussian ensembles for $V(x) = x^2/2$, and we have seen the connection between (2.6) and the one-dimensional Coulomb gas in Section 2.2. A natural extension to take here is to consider $\lambda = (\lambda_1, \dots, \lambda_N)$ as N points in $\mathbb{R}^2 \cong \mathbb{C}$, i.e. as a two dimensional Coulomb gas. Now no ordering is imposed on the eigenvalues and consequently the Hamiltonian is modified to take the modulus. Consider the example of a quadratic potential $V(z) = |z|^2$ and $\beta = 2$. Then

$$H_N(\lambda) = \frac{1}{2} \sum_{j=1}^N |\lambda_j|^2 - \frac{1}{N} \sum_{j < k} \log |\lambda_k - \lambda_j|$$

and

$$\rho(\lambda_1, \dots, \lambda_N) \propto \exp(-2N H_N(\lambda)), \quad \lambda_j \in \mathbb{C}, \quad j = 1, \dots, N$$

is the density for the eigenvalue jpdf of the complex Ginibre ensemble [61], sometimes referred to as the Ginibre Unitary Ensemble (GinUE). We can draw a parallel to the Gaussian ensembles here in the sense that the complex Ginibre ensemble comprises matrices with entries distributed according to the complex standard normal distribution, but here the symmetry constraint is dropped and hence the eigenvalues live in $\mathbb{C}$. To follow Section 2.1, an $N \times N$ matrix M from the complex Ginibre ensemble is constructed according to

$$M_{k\ell} \sim \mathcal{N}(0, \sigma^2/2) + i\mathcal{N}(0, \sigma^2/2), \quad k, \ell = 1, \dots, N. \quad (2.29)$$

When $\sigma^2 = 1/N$ the eigenvalues concentrate on the unit disk. Indeed, in the limit the eigenvalue density is known to converge to the circular law

$$\lim_{N \rightarrow \infty} \mu_N(\lambda) = \frac{1}{\pi} \mathbb{I}(|\lambda| \leq 1), \quad (2.30)$$

an analogue of the semicircle law (2.11). Here μ_N is the empirical density of eigenvalues as in (2.10). We illustrate the circular law in Figure 6 by constructing a 5000×5000 matrix according to (2.29) and plotting its eigenvalues in the complex plane.

It is very important to stress that, unlike the Gaussian ensembles, the complex Ginibre ensemble does not easily generalise to the cases $\beta \in \{1, 4\}$. Such choices correspond to the real and symplectic Ginibre ensembles (GinOE, GinSE) respectively, and the joint eigenvalue densities for these ensembles are very different and far less tractable than the complex case [16]. Indeed, only recently has a matrix model for general β been constructed [63]. These difficulties highlight an advantage of the Coulomb gas perspective, for there one can generalise $\beta > 0$ as desired (see [70] for analysis of the relationship between this two-dimensional Coulomb gas and the Ginibre ensembles). Despite this, we note that the confinement potential is still restricted to the quadratic case $V(z) = |z|^2$.

There are other variations of the β -ensembles (2.6) for which λ remains one dimensional. A simple case considers λ restrained to the unit circle. Consider $\lambda_k = e^{i\theta_k}$ with

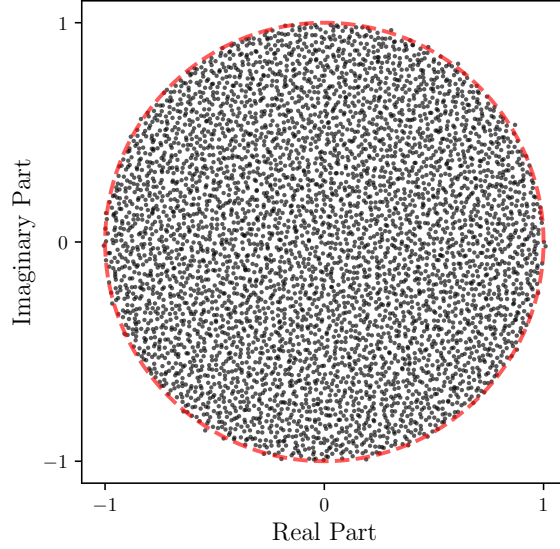


Figure 6: Eigenvalues of a 5000×5000 matrix from the complex Ginibre ensemble plotted on the complex plane. The unit disk $|z| = 1$ is superimposed in red.

$0 \leq \theta_k < 2\pi$ for $k = 1, \dots, N$. Then the joint density

$$\rho(\theta_1, \dots, \theta_N) \propto \prod_{j < k} |e^{i\theta_k} - e^{i\theta_j}|^\beta \quad (2.31)$$

describes the distribution of λ on the unit circle, and is precisely the jpdf for the eigenvalues of the circular ensembles. Notice that this form is almost identical to that seen for the β -ensembles, e.g. (2.3), simply with the confining potential removed - it is no longer needed because the domain is restricted.

The final ensembles we consider in this project are the Wishart-Laguerre ensembles. Consider an $N \times M$ matrix X comprised of independent standard normal entries over the reals, complexes, or quaternions. Then $W = \frac{1}{M} X X^\dagger$ is known as a Wishart matrix and is typically understood as a covariance matrix. By defining the aspect ratio $c := N/M$ and considering the confinement potential

$$V(x) = \frac{x}{c} - \left(\frac{1}{c} - 1 \right) \ln(x), \quad 0 < c \leq 1, \quad (2.32)$$

it so happens that the β -ensembles (2.6) precisely describe the eigenvalue jpdf for W . Again, $\beta \in \{1, 2, 4\}$ correspond to the real, complex, and symplectic cases. If one generalises to any $\beta > 0$ the ensembles are known as the β -Laguerre ensembles, named for the Laguerre polynomials that take the role of the Hermite polynomials seen in Section 2.3; indeed, they are orthogonal with respect to $e^{-V(x)}$ for V as in (2.32).

The aspect ratio c is an important parameter for the Wishart-Laguerre ensembles, for the eigenvalues concentrate in $[(1 - \sqrt{c})^2, (1 + \sqrt{c})^2] =: [a, b]$, and the limiting eigenvalue density is described by the Marčenko-Pastur law [60]

$$\lim_{N \rightarrow \infty} \mu_N(\lambda) = \frac{1}{2\pi c \lambda} \sqrt{(\lambda - a)(b - \lambda)} \mathbb{I}(a \leq \lambda \leq b). \quad (2.33)$$

More technically, (2.33) describes the limiting density for $0 < c \leq 1$, i.e. $N \leq M$. When $c > 1$ a point mass is added at zero to account for the $N - M$ trivial zero eigenvalues.

A close relative of the Marčenko-Pastur law is the quarter-circle law. This appears for the particular case of $c = 1$ when one looks at the singular values of X instead of the

eigenvalues of W . Indeed, let $\sigma = (\sigma_1, \dots, \sigma_N)$ denote the singular values of $1/\sqrt{M} \cdot X$ and note $\lambda_j = \sigma_j^2$. The support for (2.33) is on $[0, 4]$ and so, by applying the change of variables $d\lambda = 2\sigma d\sigma$, the law becomes

$$\lim_{N \rightarrow \infty} \mu_N(\sigma) = \frac{1}{\pi} \sqrt{4 - \sigma^2} \mathbb{I}(0 \leq \sigma \leq 2). \quad (2.34)$$

An illustration of both limiting densities is given in Figure 7.

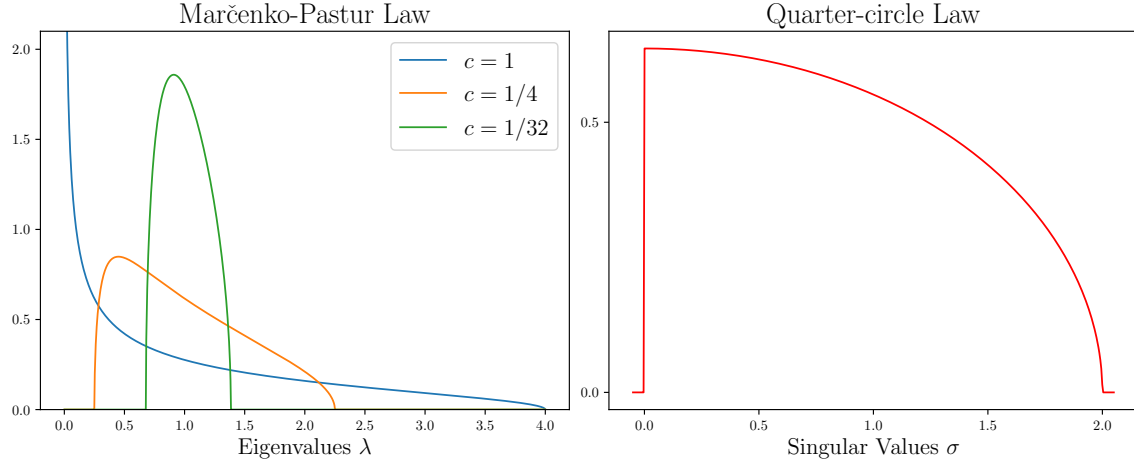


Figure 7: The limiting distribution for Wishart matrices. Left: the Marčenko-Pastur law for the eigenvalues, $c \in \{1, 1/2, 1/32\}$; right: the quarter-circle law for the singular values for the case $c = 1$.

2.7 Some Applications

Random matrix theory finds a number of applications in a surprisingly large number of fields. For example, it has seen use in physics [72], biology [59], finance [13], number theory [55], and information theory [92], to name just a few. See also the surveys [28, 53] and the references therein. In this section we explore some of these applications.

The first topic we discuss here is the problem of determining signal from noise. The idea is this: if a matrix of data, e.g. a covariance matrix, has its eigenvalues follow known laws from random matrix theory, then one may conclude that there is no correlation in the data at all - that the data is all noise. For example, recall that Wishart matrices are constructed as $W = \frac{1}{M} X X^\dagger$, where X is an $N \times M$ matrix with entries independently drawn from the standard normal distribution. They are typically seen as covariance matrices, and $c = N/M$ dictates the support of the eigenvalues $[(1 - \sqrt{c})^2, (1 + \sqrt{c})^2] =: [a, b]$. Suppose one is working with data and obtains a sample correlation matrix. If the eigenvalues of this matrix all fall within $[a, b]$ then they may reasonably assert that there is no correlation in the data. Conversely, an eigenvalue straying significantly beyond this interval would indicate a genuine correlation (signal). Analysis of stray eigenvalues is detailed in [3].

Signal detection relies on the bulk statistics of the spectrum, but we've seen that random matrix theory also makes predictions about statistics on the finer level. A humorous example concerns the distances between parked cars and between birds on a wire [82]. This study collected data for both events in a city in the Czech Republic and, after normalising the distances, observed that their spacing statistics fit the same curve. We recognise it as closely following the curve governing the spacing distribution of eigenvalues from random matrices. The phenomenon was suggested by the original authors to be the consequence of a universal distance-controlling mechanism present in both birds and humans which, as it is understood in random matrix theory, may be associated with level repulsion.

Another result relates edge fluctuation results from random matrix theory and a classical problem of combinatorics: given a uniformly random permutation of integers $\{1, \dots, N\}$, how long is its longest increasing subsequence? We let ℓ_N denote the length of longest such subsequence and define the scaled limit

$$c := \lim_{N \rightarrow \infty} \frac{1}{\sqrt{N}} \mathbb{E}(\ell_N).$$

An early contribution to the problem showed that $c = 2$ [95], which one may also recognise as the limit of the maximal eigenvalue of the Gaussian ensembles. More interestingly, it was later shown that the fluctuation around this limit as $N \rightarrow \infty$ is, again after suitable rescaling and normalisation, governed by the Tracy-Widom distribution F_2 [4].

Finally, a particularly attractive application of random matrix theory concerns the Riemann zeta function, defined

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \frac{1}{1^s} + \frac{1}{2^s} + \dots$$

when $\operatorname{Re}(s) > 1$, and through its analytic continuation elsewhere. It is the zeros of ζ that are of primary interest. By considering its functional equation [52, Chapter 5]

$$\zeta(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s) \zeta(1-s),$$

one can observe that $\zeta(-2n) = 0$ for all $n \in \mathbb{N}$. These are the ‘trivial’ zeros of the zeta function. Simply put, the Riemann hypothesis states that all non-trivial zeros lie on the critical line $s = 1/2 + it$, $t \in \mathbb{R}$.

A vast amount of work has been completed to numerically verify the Riemann hypothesis, mostly headed by Andrew Odlyzko; see, for instance, [68, 67, 34]. An interesting relationship to random matrix theory appears when looking at the statistical properties of the zeros. In particular, one can consider the mean spacings between the zeros after normalising the average spacing to be one, just as is done when studying the spacings between the eigenvalues of random matrices. To do so, we need the fact that at some point $1/2 + iT$ on the critical line, the local density of zeros is given by [88, Chapter 9]

$$\frac{1}{2\pi} \log\left(\frac{T}{2\pi}\right) + \mathcal{O}\left(\frac{1}{T^2}\right).$$

Thus if $\{\gamma_j\}_{j=1}^N$ is the set of the first N non-trivial zeros, the gaps can be normalised so that

$$\delta(\gamma_j) = \frac{1}{2\pi} \log\left(\frac{\gamma_j}{2\pi}\right) (\gamma_{j+1} - \gamma_j). \quad (2.35)$$

For an illustration we use a set of ten thousand known zeros of the zeta function starting from index $j = 10^{12} + 1$, courtesy of Odlyzko [67]. We normalise the spacings between the zeros according to (2.35) and plot a histogram of these spacings in Figure 8. Superimposed is the curve described by Wigner’s surmise for the case $\beta = 2$. There is a striking agreement between the two! Indeed, Montgomery’s pair correlation conjecture [64] asserts that, after normalisation, the pair correlation of the spacings s between any two zeros is described by

$$1 - \left(\frac{\sin(\pi s)}{\pi s}\right)^2.$$

This is precisely the pair correlation function for eigenvalues from the GUE in the bulk scaling limit. Indeed, recall the k -point correlation functions described in Section 2.3. If

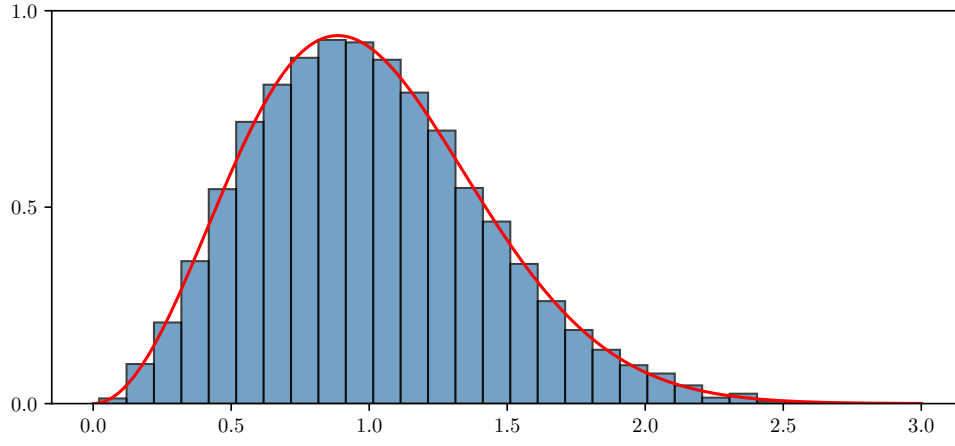


Figure 8: A histogram of the normalised spacings between zeros number $10^{12} + 1$ and $10^{12} + 10^4$ of the Riemann zeta function. The red curve is Wigner's surmise for $\beta = 2$.

$s = \lambda_2 - \lambda_1$ denotes the suitably normalised eigenvalue spacing, then [33]

$$\lim_{N \rightarrow \infty} \rho_2(\lambda_1, \lambda_2) = 1 - \left(\frac{\sin(\pi s)}{\pi s} \right)^2.$$

Montgomery's pair correlation conjecture is supported by large-scale computations by Odlyzko [67]. More generally, we recommend the work of Keating and Snaith [55] for the relationship between random matrix theory and the zeta function.

3 Sampling Schemes and Dyson Brownian Motion

In this chapter we pivot to the discussion of sampling schemes with Dyson Brownian motion (1.2). Here we write the SDE in the simpler form

$$d\lambda = -\nabla H_N(\lambda) + \sigma dW, \quad (3.1)$$

recalling that H_N is the Hamiltonian

$$H_N(\lambda) = \frac{1}{2} \sum_{j=1}^N V(\lambda_j) - \frac{1}{N} \sum_{j < k} \log(\lambda_k - \lambda_j),$$

V is the confining potential, σ is the noise scale $\sigma \equiv \sigma(\beta, N) = \sqrt{2/\beta N}$, and λ exists in the Weyl chamber $\mathbb{R}_{>}^N = \{\lambda \in \mathbb{R}^N : \lambda_1 < \dots < \lambda_N\}$. More generally, the Hamiltonian can be seen as the energy of an interacting particle system (Section 2.2) and, importantly, coincides with the density governing the β -ensembles (2.6). Thus Dyson Brownian motion is one approach for sampling the eigenvalues of random matrices. It is important to clarify that it is certainly not the only approach. Perhaps the most famous alternative scheme is the tridiagonal random matrix model [25], but in this scheme the potential is limited to the classical cases corresponding to the Gaussian and Wishart-Laguerre ensembles,

$$V(x) = \frac{x^2}{2}, \quad V(x) = \frac{x}{c} - \left(\frac{1}{c} - 1\right) \ln(x), \quad 0 < c < 1.$$

Several extensions improve on this, e.g. [36, 63], but still they are limited to at most polynomial potentials. An approach by Olver et. al. [69] exploits the determinantal structure of (2.6), as explored in Section 2.3, but naturally this scheme is limited to $\beta = 2$. Our motivation to use Dyson Brownian motion, as also employed in [57, 18], is thus threefold. First, we regain full generality of both β and the potential V . Second, its computational cost is competitive; a full comparison and more detail on the aforementioned schemes is given in Section 3.7. Third, and not least, Dyson Brownian motion poses an interesting sampling challenge in its own right.

While our main concern is the sampling of eigenvalues according to (2.6), we briefly note that there exist efficient schemes for sampling matrices from the orthogonal group [85] and the unitary group [62]. In these cases ($\beta \in \{1, 2\}$) one can construct the full matrix for the respective invariant ensembles with the transformations

$$M = O\Lambda O^T, \quad M = U\Lambda U^*,$$

where O is an orthogonal matrix, U is a unitary matrix, and Λ is the diagonal matrix of eigenvalues, $\Lambda = \text{diag}(\lambda)$. We are not aware of an efficient scheme for sampling from the symplectic group, i.e. for the case $\beta = 4$.

3.1 Euler Methods

A simple approach to the discretisation of the dynamics (3.1) is to use the Euler-Maruyama scheme

$$\lambda^{(n+1)} = \lambda^{(n)} - \Delta t \nabla H_N(\lambda^{(n)}) + \sigma \sqrt{\Delta t} \xi^{(n)}. \quad (3.2)$$

That is,

$$\lambda_k^{(n+1)} = \lambda_k^{(n)} + \left(\frac{1}{N} \sum_{j \neq k} \frac{1}{\lambda_k^{(n)} - \lambda_j^{(n)}} - \frac{1}{2} V'(\lambda_k^{(n)}) \right) \Delta t + \sigma \sqrt{\Delta t} \xi_k^{(n)}. \quad (3.3)$$

Here $\Delta t > 0$ denotes a fixed step size and $\xi^{(n)}$ is a vector of standard normal random variables in $\mathbb{R}^N$. The Euler-Maruyama scheme is popular because it is simple and inexpensive to implement. However, the scheme requires the vector field $\nabla H_N(\lambda)$ to be globally Lipschitz in order for it to be well posed [46, 47]. This is certainly not the case here because both the Coulomb interaction

$$\frac{1}{N} \sum_{j \neq k} \frac{1}{\lambda_k - \lambda_j}$$

and any potential whose derivative is superlinear at infinity, e.g. the quartic case $V(x) = x^4/4$, violate the Lipschitz assumption.

A standard method to tackle problems with superlinear drift is to use a ‘tamed’ explicit scheme [51]. Indeed, this is the approach of the authors in [57] who write the explicit update to (3.1) as

$$\lambda_k^{(n+1)} = \lambda_k^{(n)} + \left(\frac{1}{N} \sum_{j \neq k} \frac{1}{\lambda_k^{(n)} - \lambda_j^{(n)}} - \frac{V'(\lambda_k^{(n)})}{2 + \Delta t |V'(\lambda_k^{(n)})|} \right) \Delta t + \sigma \sqrt{\Delta t} \xi_k^{(n)}. \quad (3.4)$$

Notice that here only the potential V has been tamed, and it is thus possible to see the update as a modification of the standard Euler scheme with a second order correction,

$$\lambda_k^{(n+1)} = \lambda_k^{(n)} - \Delta t \nabla H_N(\lambda) + \sigma \sqrt{\Delta t} \xi_k^{(n)} + \frac{V'(\lambda_k^{(n)}) |V'(\lambda_k^{(n)})|}{4 + 2\Delta t |V'(\lambda_k^{(n)})|} (\Delta t)^2.$$

The continued presence of the Coulomb interaction means (3.4) remains ill-conditioned and does not satisfy the assumptions governing the convergence of tamed schemes outlined in [51]. Thus, while we may expect the scheme to perform better than the standard scheme (3.2), particularly when V' is superlinear at infinity, it will likely meet the same problems concerning stability and convergence. Perhaps the most important problem on this theme is the possibility that any two eigenvalues have their paths intersect; indeed, it is shown in [65] that the explicit scheme fails to prevent the trajectories of two eigenvalues from crossing. The importance of this detail is explored further in Section 3.2.

Finally, we also consider discretising the Langevin dynamics (3.1) with an implicit Euler step,

$$\lambda^{(n+1)} = \lambda^{(n)} - \Delta t \nabla H_N(\lambda^{(n+1)}) + \sigma \sqrt{\Delta t} \xi^{(n)} \quad (3.5)$$

by considering

$$\begin{aligned} \lambda^{(n+1)} &= \text{prox}_{\Delta t, H_N}(\lambda^{(n)} + \sigma \sqrt{\Delta t} \xi^{(n)}) \\ &= \arg \min_x \left(H_N(x) + \frac{1}{2\Delta t} \|x - \lambda^{(n)} - \sigma \sqrt{\Delta t} \xi^{(n)}\|_2^2 \right) =: \arg \min_x g(x). \end{aligned} \quad (3.6)$$

Here $\text{prox}_{c,f}$ indicates the proximal operator [71]

$$\text{prox}_{c,f}(v) := \arg \min_x \left(f(x) + \frac{1}{2c} \|x - v\|_2^2 \right), \quad (3.7)$$

and one can see the equivalence between (3.5) and (3.6) by adopting the convenient definition of $g(x)$ as in (3.6) and observing

$$\begin{aligned} \nabla_x g(x) = 0 &\implies \nabla H_N(x) + \frac{1}{\Delta t} (x - \lambda^{(n)} - \sigma \sqrt{\Delta t} \xi^{(n)}) = 0 \\ &\implies x = \lambda^{(n)} - \Delta t \nabla H_N(x) + \sigma \sqrt{\Delta t} \xi^{(n)} \end{aligned}$$

where $x := \lambda^{(n+1)}$. The form (3.6) is immediately useful because it makes it clear that

no eigenvalue crossings can happen with the implicit Euler scheme. Indeed, the energy H_N is infinite for any $\lambda \notin \mathbb{R}_{>}^N$ and therefore such cases will never solve the minimisation problem. We provide some analysis of the scheme here. First consider the gradient

$$[\nabla H_N(x)]_k = \frac{1}{2}V'(x_k) - \frac{1}{N} \sum_{j \neq k} \frac{1}{x_k - x_j},$$

and then observe that the Hessian can be written

$$[\nabla^2 H_N(x)]_{k\ell} = \begin{cases} -\frac{1}{N(x_k - x_\ell)^2}, & k \neq \ell, \\ \frac{1}{2}V''(x_k) + \frac{1}{N} \sum_{j \neq k} \frac{1}{(x_k - x_j)^2}, & k = \ell. \end{cases} \quad (3.8)$$

Recall that a matrix A is strictly diagonally dominant (SDD) if for all $i = 1, \dots, N$

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}|.$$

It is clear from the somewhat trivial inequality

$$\left| \frac{1}{2}V''(x_k) + \frac{1}{N} \sum_{j \neq k} \frac{1}{(x_k - x_j)^2} \right| > \frac{1}{N} \sum_{j \neq k} \frac{1}{(x_k - x_j)^2}$$

that the Hessian (3.8) is SDD for all strictly convex potentials ($V''(x) > 0$), and therefore is positive definite (e.g. by appealing to Gershgorin's circle theorem). In these cases H_N is therefore convex. In fact, for the minimum of g as defined in (3.6) we have

$$\nabla^2 g(x) = \nabla^2 H_N(x) + \frac{1}{\Delta t} I_N,$$

where I_N is the $N \times N$ identity matrix. Therefore g is strongly convex in the sense that $\nabla^2 g(x) \succeq 1/\Delta t \cdot I_N \succ 0$. By extending the definition of g so that

$$g(x) = \begin{cases} H_N(x) + \frac{1}{2\Delta t} \left\| x - \lambda^{(n)} - \sigma \sqrt{\Delta t} \xi^{(n)} \right\|_2^2, & x \in \mathbb{R}_{>}^N, \\ +\infty, & \text{otherwise,} \end{cases}$$

it follows that g is a proper closed convex function and that there is a unique minimiser to (3.6) for all $v = \lambda^{(n)} + \sigma \sqrt{\Delta t} \xi^{(n)}$ [71]. Furthermore, the damped Newton method for root finding

$$x^{(m+1)} = x^{(m)} - \alpha^{(m)} [\nabla^2 g(x^{(m)})]^{-1} \nabla g(x^{(m)}) \quad (3.9)$$

converges to this unique minimiser [66]. That is, we implement backtracking for the Newton step parameter $\alpha^{(m)}$ to ensure $x^{(m+1)}$ remains ordered, i.e. $x^{(m+1)} \in \mathbb{R}_{>}^N$, and that a sufficient decrease condition is met. Additionally, we use the conjugate gradient method for the inverse Hessian

$$[\nabla^2 g(x^{(m)})]^{-1}. \quad (3.10)$$

This is an iterative method that finds an approximation to (3.10), the main appeal being that it is of $\mathcal{O}(N^2)$ complexity [91] instead of $\mathcal{O}(N^3)$, the complexity of a naive inverse.

To conclude our discussion of the implicit scheme we comment on its convergence properties. The implicit Euler scheme is known to converge strongly to the solution for nonlinear stochastic differential equations whose drift $b : \mathbb{R}^N \rightarrow \mathbb{R}^N$ satisfies a one-sided Lipschitz condition [47]. Now, the vector field $-\nabla H_N$ is indeed one-sided Lipschitz on the Weyl chamber $\mathbb{R}_{>}^N$. This is a direct consequence of the convexity of H_N , for then ∇H_N is

monotone and

$$\langle \lambda_1 - \lambda_2, \nabla H_N(\lambda_1) - \nabla H_N(\lambda_2) \rangle \geq 0$$

for all $\lambda_1, \lambda_2 \in \mathbb{R}_{>}^N$. It immediately follows that

$$\langle \lambda_1 - \lambda_2, (-\nabla H_N(\lambda_1)) - (-\nabla H_N(\lambda_2)) \rangle \leq 0$$

for all $\lambda_1, \lambda_2 \in \mathbb{R}_{>}^N$, i.e. the vector field is one-sided Lipschitz with Lipschitz constant zero. The problem here is that classical convergence analysis (e.g. [47]) concerns functions defined on $\mathbb{R}^N$. To our knowledge analysis for domain-restricted SDEs is limited. The particular case of Dyson Brownian motion and the Weyl chamber does fall under a more general framework [65, 24], but the assumptions in these cases would demand unreasonable restrictions on β . Thus we choose to leave the problem of the convergence rate for the implicit scheme open. What we can say is that it is well posed for every $\Delta t > 0$, and we will attempt to measure the empirical convergence of the scheme in Section 4.3.

3.2 Intersections and the Energy Landscape

Discussion surrounding the Euler methods in Section 3.1 proposed that the discretisation scheme should respect the non-colliding nature of (3.1); in other words, that λ remains in the Weyl chamber $\mathbb{R}_{>}^N = \{\lambda \in \mathbb{R}^N : \lambda_1 < \dots < \lambda_N\}$ across the whole trajectory. In this section we explore why this is important and discuss the landscape of Dyson Brownian motion for the purpose of analysing the implicit update (3.5).

The first point we discuss here concerns the existence and uniqueness of solutions to (3.1). This is a very well studied area; see, for instance, [76, 40, 17, 41] and [1, Chapter 4]. In essence, assuming (a) the initial position $\lambda(0) \in \mathbb{R}_{>}^N$, (b) that V is well behaved (convexity suffices), and (c) that $\beta \geq 1$, then there is a unique solution to (3.1) satisfying $\lambda(t) \in \mathbb{R}_{>}^N$ almost surely for any $t > 0$. Thus to remain faithful to the continuous time system the discretisation must also respect the non-colliding nature of the eigenvalues. Moreover, schemes with no intersections can be used for analysis of more than just the macroscopic density (2.6). Any scheme admitting trajectories that intersect is certainly not well suited for analysis of the spacing distribution! It is also plausible that eigenvalue intersection adversely affects the spectrum's edge fluctuation.

Let us now revisit the explicit Euler scheme (3.3) to understand why eigenvalue crossings occur. For simpler analysis we consider the case $V(x) = x^2/2$ and suppose there exists a pair of eigenvalues whose gap is small relative to the distance from any other eigenvalue. Formally, the pair $(\lambda_\ell, \lambda_m)$, with $\lambda_\ell < \lambda_m$, has $\delta := \lambda_m - \lambda_\ell > 0$ such that

$$\delta \ll \eta := \min_{k \neq \ell, m} (|\lambda_k - \lambda_\ell|, |\lambda_k - \lambda_m|). \quad (3.11)$$

In this regime we decouple the singular repulsion force from the interaction with other eigenvalues and the confining potential. Denoting the drift of (3.1) as $b(\lambda)$,

$$\begin{aligned} b(\lambda_\ell) &= \frac{1}{N} \cdot \frac{1}{\lambda_\ell - \lambda_m} + \frac{1}{N} \sum_{j \neq \ell, m} \frac{1}{\lambda_\ell - \lambda_j} - \frac{1}{2} V'(\lambda_\ell) =: -\frac{1}{N\delta} + F_\ell, \\ b(\lambda_m) &= \frac{1}{N} \cdot \frac{1}{\lambda_m - \lambda_\ell} + \frac{1}{N} \sum_{j \neq m, \ell} \frac{1}{\lambda_m - \lambda_j} - \frac{1}{2} V'(\lambda_m) =: \frac{1}{N\delta} + F_m. \end{aligned}$$

It is convenient to write $F_\ell = G(\lambda_\ell)$ and $F_m = G(\lambda_m)$ where

$$G(x) = \frac{1}{N} \sum_{j \neq \ell, m} \frac{1}{x - \lambda_j} - \frac{1}{2} V'(x).$$

Notice that G is differentiable on $[\lambda_\ell, \lambda_m]$. By the mean value theorem, there exists a

point $c \in (\lambda_\ell, \lambda_m)$ such that

$$F_m - F_\ell = G(\lambda_m) - G(\lambda_\ell) = G'(c)(\lambda_m - \lambda_\ell) =: G'(c)\delta = \mathcal{O}(\delta). \quad (3.12)$$

Hence $F_m - F_\ell$ vanishes to leading order as $\delta \rightarrow 0$. This difference is dropped in the approximation $b(\lambda_m) - b(\lambda_\ell) \approx 2/(N\delta)$ and, since

$$d\lambda_m = b(\lambda_m)dt + \sigma dW_m, \quad d\lambda_\ell = b(\lambda_\ell)dt + \sigma dW_\ell, \quad \sigma \equiv \sqrt{2/(\beta N)},$$

where dW_m, dW_ℓ are independent Brownian motions, we find

$$d\delta = d\lambda_m - d\lambda_\ell = \frac{2}{N\delta}dt + \sqrt{2}\sigma dW \quad (3.13)$$

using the independence of Brownian motion, $dW_m - dW_\ell = \sqrt{2}dW$. The Euler-Maruyama discretisation for (3.13) is

$$\delta^{(n+1)} = \delta^{(n)} + \Delta t \frac{2}{N\delta^{(n)}} + \sqrt{\frac{4\Delta t}{\beta N}} \xi^{(n)}$$

for $\xi^{(n)} \sim \mathcal{N}(0, 1)$ and $\Delta t > 0$. Therefore a crossing occurs ($\delta^{(n+1)} < 0$) when

$$\xi^{(n)} < -\sqrt{\frac{\beta N}{4\Delta t}} \left(\delta^{(n)} + \frac{2\Delta t}{N\delta^{(n)}} \right). \quad (3.14)$$

A simple estimate for N eigenvalues in a fixed interval suggests $\delta \sim 1/N$. However, it is known that for the Gaussian ensembles with $\beta = 2$, for example, the minimum eigenvalue spacing scales like $\delta^{(n)} \sim N^{-4/3}$ [7]. Therefore in this case (3.14) becomes

$$\xi^{(n)} < -\sqrt{\frac{N}{2\Delta t}} \left(\frac{1}{N^{4/3}} + 2\Delta t N^{1/3} \right) = -\left(\frac{1}{N^{5/6}\sqrt{2\Delta t}} + N^{5/6}\sqrt{2\Delta t} \right). \quad (3.15)$$

We continue the analysis of (3.15). Let Φ denote the standard normal cumulative distribution function,

$$\Phi(x) = \mathbb{P}(\xi \leq x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt, \quad \xi \sim \mathcal{N}(0, 1),$$

and define $p = N^{5/6}\sqrt{2\Delta t}$ and $q = 1/p$. Notice then that (3.15) becomes $\xi^{(n)} < -(p + q)$ and therefore the probability of a crossing is $\Phi(-(p + q))$. Since $\Phi(-x)$ is decreasing, a smaller crossing probability implies $p + q$ must increase. Let ϵ denote a small probability that a crossing occurs and define z such that

$$\Phi(-(p + q)) \leq \epsilon \implies p + q \geq -\Phi^{-1}(\epsilon) = \Phi^{-1}(1 - \epsilon) =: z. \quad (3.16)$$

Here we have used the symmetry $\Phi(-x) = 1 - \Phi(x)$. To solve $p + q \geq z$ we need

$$p + \frac{1}{p} \geq z \implies p^2 - zp + 1 \geq 0 \quad (3.17)$$

and therefore there are two roots for p ,

$$p_- = \frac{z - \sqrt{z^2 - 4}}{2}, \quad p_+ = \frac{z + \sqrt{z^2 - 4}}{2}, \quad (3.18)$$

with $\{p : p \leq p_-\}$ and $\{p : p \geq p_+\}$ satisfying the inequality (3.17). Recall $p = N^{5/6}\sqrt{2\Delta t}$.

For an example, take the lower branch p_- and consider $\epsilon = 10^{-4}$. Then

$$p = N^{5/6} \sqrt{2\Delta t} \leq p_- \implies \Delta t \leq \frac{p_-^2}{2} N^{-5/3},$$

and approximating $z = \Phi^{-1}(1 - 10^{-4}) \approx 3.719$, we have $\Delta t \lesssim 0.04 \cdot N^{-5/3}$ as an estimate for the step size needed for no crossings to occur with the explicit Euler scheme with probability $1 - 10^{-4}$. More generally, this analysis suggests $\Delta t \sim \mathcal{O}(N^{-5/3})$ is the required scaling for no crossings to occur.

It is important to address some nuances here. We emphasise that this analysis is for an idealised two body problem in the pair $(\lambda_\ell, \lambda_m)$. A key assumption was that we can treat the interaction between other eigenvalues, together with the confining potential, as background noise; that is, the difference $F_m - F_\ell$ in (3.12) is $\mathcal{O}(\delta)$ and dominated by the singular term $2/N\delta$, thus dropped to achieve (3.13). But, for this assumption to remain valid, the motion of $(\lambda_\ell, \lambda_m)$ must be confined to a neighbourhood for which $\delta \ll \eta$. If Δt is too large, such as in the positive branch cut $p > p_+$, this would be violated and a crossing would occur between one of λ_ℓ, λ_m and a third previously distant eigenvalue. Hence we consider only the smaller branch $p < p_-$. Numerical experiments for this analysis are provided in Section 4.2.

3.3 Metropolis-Hastings Algorithms

It is well known that unadjusted Langevin algorithms introduce an inherent discretisation bias: that they converge to a biased stationary distribution $\pi_{\Delta t}(x)$ instead of the true target density $\pi(x) \propto e^{-U(x)}$. A popular approach to address this bias is the introduction of a Metropolis-Hastings acceptance step. If one considers a proposal from the general ‘stochastic theta method’ [45]

$$\lambda^{(n+1)} = \lambda^{(n)} - \Delta t(1 - \theta)\nabla H_N(\lambda^{(n)}) - \Delta t\theta\nabla H_N(\lambda^{(n+1)}) + \sigma\sqrt{\Delta t}\xi^{(n)}, \quad (3.19)$$

then, writing $y = \lambda^{(n+1)}$ for the proposed step and $x = \lambda^{(n)}$ for the current state, the proposal is accepted or rejected according to the probability

$$\alpha(x, y) = \min \left\{ 1, \frac{\pi(y)q(y, x)}{\pi(x)q(x, y)} \right\}. \quad (3.20)$$

As discussed in [49], the transition kernel $q(x, y)$ for the standard Langevin dynamics $d\lambda = -\nabla U(\lambda) + \sqrt{2}dW$ is determined by

$$\frac{|\det(I_N + \Delta t\theta\nabla^2 U(y))|}{(4\pi\Delta t)^{N/2}} \exp \left(-\frac{1}{4\Delta t} \|y - x + \Delta t(\theta\nabla U(y) + (1-\theta)\nabla U(x))\|_2^2 \right).$$

However, since our dynamics have $U(x) = \beta N H_N(x)$ and variance $\sigma = \sqrt{2/\beta N}$, the transition kernel is instead

$$q(x, y) = \frac{|\det(I_N + \Delta t\theta\nabla^2 H_N(y))|}{(4\pi\Delta t/(\beta N))^{N/2}} \exp \left(-\frac{\beta N}{4\Delta t} \|y - x + \Delta t(\theta\nabla H_N(y) + (1-\theta)\nabla H_N(x))\|_2^2 \right). \quad (3.21)$$

It is also common to express $q(x, y)$ in a way that makes its Gaussian nature more clear. If $\phi(z; \mu, \Sigma)$ denotes the density of a Gaussian evaluated at z with mean μ and variance Σ , then (3.21) is equivalent to

$$q(x, y) = |\det(I_N + \Delta t\theta\nabla^2 H_N(y))| \phi \left(y + \Delta t\theta\nabla H_N(y); x - \Delta t(1-\theta)\nabla H_N(x), \frac{2\Delta t}{\beta N} I \right). \quad (3.22)$$

The proposal (3.19) corresponds to the explicit and implicit Euler methods described in Chapter 3.1 when $\theta = 0$ and $\theta = 1$ respectively. Indeed, $\theta = 0$ recovers the common Metropolis adjusted Langevin algorithm (MALA). When $\theta = 1$ the algorithm is known as the implicit Langevin algorithm (ILA), for which the need to calculate the ratio of determinants

$$\frac{|\det(I_N + \Delta t \theta \nabla^2 H_N(x))|}{|\det(I_N + \Delta t \theta \nabla^2 H_N(y))|} \quad (3.23)$$

arises (and in general appears for any $\theta > 0$). This is an expensive procedure to carry out numerically, typically of order at least $\mathcal{O}(N^3)$ [91]. Thus MALA is a natural choice. Despite this, we consider implicit schemes in our experiments and discuss algorithms that address (3.23) in Section 3.6.

The form (3.19) is the classical discretisation for a one-step algorithm. However, there is an alternative characterisation known as an implicit midpoint step,

$$\lambda^{(n+1)} = \lambda^{(n)} - \Delta t \nabla H_N(\theta \lambda^{(n+1)} + (1 - \theta) \lambda^{(n)}) + \sigma \sqrt{\Delta t} \xi^{(n)}, \quad (3.24)$$

well described in [56]. As was shown for the implicit Euler step (3.5), for $\theta > 0$ we can see (3.24) in terms of the proximal operator (3.7),

$$\lambda^{(n+1)} = \left(1 - \frac{1}{\theta}\right) \lambda^{(n)} + \frac{1}{\theta} \text{prox}_{\theta \Delta t, H_N}(\lambda^{(n)} + \theta \sigma \sqrt{\Delta t} \xi^{(n)}). \quad (3.25)$$

It is possible to extend (3.25) so that it too can be written as a solution to a minimisation problem, albeit with some more work than before. We follow [56] whose argument uses the fact that if $g(x) = f(\alpha x + \beta)$ for $\alpha \neq 0$ and β a fixed constant (i.e. g is an affine transformation of x), then [71, Chapter 2.2]

$$\text{prox}_{\gamma, g}(v) = \frac{1}{\alpha} (\text{prox}_{\alpha^2 \gamma, f}(\alpha v + \beta) - \beta). \quad (3.26)$$

So, define the constants $\alpha = \theta$, $\beta = (1 - \theta) \lambda^{(n)}$, $\gamma = \Delta t / \theta$, functions $f(x) = H_N(x)$ and $g(x) = H_N(\alpha x + \beta) = H_N(\theta x + (1 - \theta) \lambda^{(n)})$, and proximal argument $v = \lambda^{(n)} + \sigma \sqrt{\Delta t} \xi^{(n)}$ so that the right hand side of (3.26) is

$$\begin{aligned} & \frac{1}{\theta} \left(\arg \min_x \left(H_N(x) + \frac{1}{2\theta \Delta t} \left\| x - \theta \lambda^{(n)} - \theta \sigma \sqrt{\Delta t} \xi^{(n)} - (1 - \theta) \lambda^{(n)} \right\|_2^2 \right) - (1 - \theta) \lambda^{(n)} \right) \\ &= \left(1 - \frac{1}{\theta} \right) \lambda^{(n)} + \frac{1}{\theta} \arg \min_x \left(H_N(x) + \frac{1}{2\theta \Delta t} \left\| x - \lambda^{(n)} - \theta \sigma \sqrt{\Delta t} \xi^{(n)} \right\|_2^2 \right) \\ &= \left(1 - \frac{1}{\theta} \right) \lambda^{(n)} + \frac{1}{\theta} \text{prox}_{\theta \Delta t, H_N}(\lambda^{(n)} + \theta \sigma \sqrt{\Delta t} \xi^{(n)}). \end{aligned}$$

That is, the right hand side of (3.26) is precisely the proximal form of the implicit midpoint step (3.25). This relation therefore gives the equivalent formulation

$$\lambda^{(n+1)} = \arg \min_x \left(H_N(\theta x + (1 - \theta) \lambda^{(n)}) + \frac{\theta}{2\Delta t} \left\| x - \lambda^{(n)} - \sigma \sqrt{\Delta t} \xi^{(n)} \right\|_2^2 \right),$$

which draws a clear parallel to the implicit Euler step (3.6).

The implicit midpoint step is of particular interest in a Metropolis-Hastings framework with $\theta = 1/2$. Indeed, again writing $y = \lambda^{(n+1)}$ for the proposal and $x = \lambda^{(n)}$ for the

current state, the transition kernel $q(x, y)$ for (3.24) is²

$$\frac{|\det(I_N + \Delta t \theta \nabla^2 H_N(\theta y + (1 - \theta)x))|}{(4\pi \Delta t / (\beta N))^{N/2}} \exp\left(-\frac{\beta N}{4\Delta t} \|y - x + \Delta t \nabla H_N(\theta y + (1 - \theta)x)\|_2^2\right). \quad (3.27)$$

Observe that when $\theta = 1/2$, the determinant is symmetric in x and y and therefore the ratio of determinants (3.23) falls out from the acceptance ratio

$$\alpha(x, y) = \min\left\{1, \frac{\pi(y)q(y, x)}{\pi(x)q(x, y)}\right\}.$$

That is, one can benefit from the improved properties of the implicit midpoint method without needing to calculate the expensive determinant ratio previously seen. We refer to algorithms with this step as the (Metropolis-Adjusted) Implicit Midpoint Langevin Algorithm (IMLA, MAIMLA).

3.4 Hamiltonian Monte Carlo

Hybrid or Hamiltonian Monte Carlo (HMC) is a Metropolis-Hastings algorithm designed to overcome the random walk-like behaviour that gradient based schemes can exhibit. The main idea of the algorithm is to introduce an auxiliary ‘momentum’ variable p and then produce proposals according to the approximation of Hamiltonian dynamics through a geometric integrator, commonly the leapfrog (Verlet) scheme. In this project we do not explore the theory concerning HMC in much detail; see instead [80, 81, 9], for example. Our focus lies on the discussion of one HMC algorithm for Dyson Brownian motion proposed by Chafaï and Ferré [18]. It incorporates friction with the parameter γ_N and a time scaling parameter α_N , but to remain consistent with their experiments we consider $\gamma_N = \alpha_N = 1$. With these choices, the algorithm as can be summarised as follows. Let $\Delta t > 0$ be the step size, $x = \lambda^{(n)}$ the current position and $p = p^{(n)}$ the current momentum, with $y = \lambda^{(n+1)}$ and $q = p^{(n+1)}$ denoting the proposals for the next step. First update the momentum,

$$\tilde{p} = e^{-\Delta t} p + \sqrt{\frac{1 - e^{-2\Delta t}}{\beta N}} \xi^{(n)},$$

where $\xi^{(n)} \sim \mathcal{N}(0, I_N)$. Then run the leapfrog integrator to produce the proposals,

$$\begin{cases} \tilde{q} = \tilde{p} - \nabla H_N(x) \Delta t / 2, \\ y = x + \tilde{q} \Delta t, \\ q = \tilde{q} - \nabla H_N(y) \Delta t / 2, \end{cases} \quad (3.28)$$

and finish by computing the probability ratio

$$\min\left\{1, \exp\left[-\beta N \left(H_N(y) + \frac{\|q\|^2}{2} - H_N(x) - \frac{\|\tilde{p}\|^2}{2}\right)\right]\right\}. \quad (3.29)$$

Accept the proposal, i.e. $(\lambda^{(n+1)}, p^{(n+1)}) = (y, q)$, according to the probability (3.29). Otherwise, update the momentum so that $p^{(n+1)} = -\tilde{p}$.

An important point to communicate here is that only one iteration of the leapfrog integrator (3.28) is applied. Typically HMC algorithms run $L > 1$ steps of the integrator and, if we denote the input step size as ε , consider the product $L\varepsilon$ together when discussing computational complexity and scaling. In fact, when $L = 1$ the proposal produced by

²From private communication.

(3.28) is nothing more than a rescaled MALA proposal

$$y = x - \Delta t_{\text{MALA}} \nabla H_N(x) + \sqrt{\frac{2\Delta t_{\text{MALA}}}{\beta N}} \xi. \quad (3.30)$$

Recall ξ is an N -dimensional standard normal random variable. For a simpler comparison, we write the MALA proposal in its more common form by introducing $U(x) = \beta N H_N(x)$ so that

$$y = x - h_{\text{MALA}} \nabla U(x) + \sqrt{2h_{\text{MALA}}} \xi, \quad h_{\text{MALA}} = \frac{\Delta t_{\text{MALA}}}{\beta N}. \quad (3.31)$$

Now reconsider the leapfrog integrator (3.28). Since only one step is taken, we can directly evaluate the update

$$\begin{aligned} y &= x + (\tilde{p} - \nabla H_N(x) \Delta t_{\text{HMC}}/2) \Delta t_{\text{HMC}} \\ &= x - \frac{(\Delta t_{\text{HMC}})^2}{2\beta N} \nabla U(x) + \Delta t_{\text{HMC}} \tilde{p}. \end{aligned}$$

The Hamiltonian for these dynamics, as apparent from the acceptance probability (3.29), is $\mathcal{H}(x, p) = \beta N (H_N(x) + \|p\|^2/2)$. Since the joint distribution $\pi(x, p) \propto \exp(-\mathcal{H}(x, p))$, we have that $\pi(p) \propto \exp(-\beta N \|p\|^2/2)$ and therefore $p \sim \mathcal{N}(0, I_N/\beta N)$. It follows that the proposal can be written

$$\begin{aligned} y &= x - \frac{(\Delta t_{\text{HMC}})^2}{2\beta N} \nabla U(x) + \frac{\Delta t_{\text{HMC}}}{\sqrt{\beta N}} \xi \\ &= x - h_{\text{HMC}} \nabla U(x) + \sqrt{2h_{\text{HMC}}} \xi, \quad h_{\text{HMC}} = \frac{(\Delta t_{\text{HMC}})^2}{2\beta N}. \end{aligned} \quad (3.32)$$

So, when only one integrator step is taken, we can directly compare the proposals (3.31) and (3.32) to see $\Delta t_{\text{HMC}} = \sqrt{2\Delta t_{\text{MALA}}}$. This relation does not imply that the one-step HMC algorithm is an improvement over MALA. Indeed, the proposals are consistent discretisations of (3.1) and the quantity that controls how much a single step advances the chain is h , not Δt . That is to say that both schemes must take the same number of steps to achieve the same effective diffusion time. To illustrate this idea with an example, consider a single realisation with dimension $N = 50$ that wants to run for an effective time of $T = 3.0$. Suppose MALA needs the scaling $\Delta t_{\text{MALA}} \sim 1/N^2$ and, accordingly, that HMC needs the scaling $\Delta t_{\text{HMC}} \sim 1/N$. Let n denote the total number of steps taken by each method. Then MALA needs $n = \lceil T/\Delta t_{\text{MALA}} \rceil = 7500$ steps to achieve an effective time of $T = 3.0$, and HMC needs this same number of steps in order to achieve the same effective time. Running only $n = \lceil T/\Delta t_{\text{HMC}} \rceil = 150$ steps would see minimal progress of the chain from its initialisation. Numerical experiments that illustrate this idea are given in Section 4.4.

3.5 Fast Multipole Methods

Of central importance in Dyson Brownian motion is what we have called the ‘Coulomb interaction’

$$\frac{1}{N} \sum_{j \neq k} \frac{1}{\lambda_k - \lambda_j}, \quad (3.33)$$

whose name comes from the log-gas framework described in Section 2.2. It is important not just because it is highly singular, but also because of its inherent $\mathcal{O}(N^2)$ complexity. Indeed, there is no avoiding the interaction and it often appears multiple times in the algorithms previously described.

One way to reduce the complexity of (3.33) is to approximate the interaction through

a ‘fast multipole method’. These are well described in the original work of Greengard and Rokhlin [42]. Informally, the algorithm separates the influence of near and far neighbours for each particle, calculating the near neighbour interaction exactly but approximating the influence of distant neighbours. Its main appeal is that its asymptotic complexity for a fixed precision is $\mathcal{O}(N)$, although common variations of the algorithm are $\mathcal{O}(N \log N)$; see the discussion in [96]. In this section we spend some time exploring how the algorithm works. The argument is quite technical and we again recommend the original work [42] but also the supplementary material [5] and [44, Chapter 5].

Consider a sorted particle system on $\mathbb{R}$, e.g. $\lambda \in \mathbb{R}_>^N = \{\lambda \in \mathbb{R}^N : \lambda_1 < \dots < \lambda_N\}$. We note that $\mathbb{R}$ is chosen as the domain only for simpler analysis, and that FMM generalises to higher dimensions easily. To begin, continuously bisect the particle support into a binary tree until each leaf has at most s particles. The parameter s influences the speed and the precision of the algorithm. We call the interval containing each subset of particles a box, B , with the centre of any box denoted c_B . Within the tree structure we may interchangeably refer to boxes as nodes. Consider any particle λ_j from a box B together with a ‘distant’ particle x . Here ‘distant’ has a precise meaning in that $|\lambda_j - c_B| < |x - c_B|$ for all $\lambda_j \in B$. We may then express the repulsion influence of all particles in B on the target x as

$$\sum_{\lambda_j \in B} \frac{1}{x - \lambda_j} = \sum_{\lambda_j \in B} \frac{1}{(x - c_B) - (\lambda_j - c_B)} = \sum_{\lambda_j \in B} \frac{1}{x - c_B} \frac{1}{1 - \frac{\lambda_j - c_B}{x - c_B}}.$$

By defining $r := (\lambda_j - c_B)/(x - c_B)$, and noting $|r| < 1$ by the distant nature of x ,

$$\sum_{\lambda_j \in B} \frac{1}{x - \lambda_j} = \sum_{\lambda_j \in B} \frac{1}{x - c_B} \frac{1}{1 - r} = \sum_{\lambda_j \in B} \frac{1}{x - c_B} \sum_{k=0}^{\infty} r^k.$$

A swap of the summations and re-substitution of r yields

$$\sum_{\lambda_j \in B} \frac{1}{x - \lambda_j} = \sum_{k=0}^{\infty} \frac{1}{(x - c_B)^{k+1}} \sum_{\lambda_j \in B} (\lambda_j - c_B)^k =: \sum_{k=0}^{\infty} \frac{M_k}{(x - c_B)^{(k+1)}}, \quad (3.34)$$

where M_k is known as the k th multipole moment. Here the infinite sum on the right hand side is truncated to the first p terms and will be called the p -multipole expansion of a box, where p can be chosen according to the desired precision of the algorithm. In sum, the p -multipole expansion is an approximation for the repulsion influence of all particles in a box on a distant target x .

While we could build the p -multipole moments M_k for every box individually, it is possible and far more efficient to construct them hierarchically by performing an upwards pass through the tree, starting at the leaves. To see this, let M_k^C be the multipole moments for some child node, M_k^P the moments for the parent, and c_P, c_C the centres of each node respectively. Notice that one can recentre

$$(\lambda_j - c_P)^k = ((\lambda_j - c_C) + (c_C - c_P))^k =: ((\lambda_j - c_C) + \delta)^k$$

with $\delta := c_C - c_P$. By standard binomial expansion

$$(\lambda_j - c_P)^k = \sum_{n=0}^k \binom{k}{n} (\lambda_j - c_C)^n \delta^{k-n}.$$

Now, every particle in the parent node belongs to exactly one of its two child nodes. If

C_L and C_R denote the (left and right) children of the parent P , notice that

$$M_k^P = \sum_{\lambda_j \in P} (\lambda_j - c_P)^k = \sum_{n=0}^k \binom{k}{n} M_n^{C_L} (c_{C_L} - c_P)^{k-n} + \sum_{n=0}^k \binom{k}{n} M_n^{C_R} (c_{C_R} - c_P)^{k-n}.$$

That is, the parent's multipole moments depend only on the multipole moments of its children. Thus we need only calculate these for the leaves of the tree before they can be propagated up to every other box. This step is the ‘multipole-to-multipole’ (M2M) stage of the algorithm and concludes the upwards pass.

So far, each box has its own multipole expansion (3.34) that describes the box's influence on any given distant particle. In order to calculate the total far-field influence on this particle, one would need to evaluate the multipole expansions for every distant box individually. The next steps in the algorithm improve upon this. First, we build one expansion at the centre of every box that collects the repulsion influence of every distant box. This is the ‘multipole-to-local’ (M2L) expansion. Formally, let $\mathcal{I}(B)$ denote the interaction list of a box, defined as the boxes on the same level as B , not adjacent to B , but whose parents are adjacent to B 's parent. Later we use the M2L expansions passing top-down through the tree, and the interaction list ensures every pair of boxes is only counted once; a box not adjacent to B is far enough for the multipole approximation to apply, but if its parent was also not adjacent to B 's parent then the pair was already accounted for on the previous tree level. Consider some particle $x \in B$, the p -multipole expansion of each $S \in \mathcal{I}(B)$, and define $R_S := (c_B - c_S)$ in order to recentre each expansion in B ,

$$\sum_{k=0}^p \frac{M_k^S}{(x - c_S)^{k+1}} = \sum_{k=0}^p \frac{M_k^S}{((x - c_B) + (c_B - c_S))^{k+1}} = \sum_{k=0}^p \frac{M_k^S}{((x - c_B) + R_S)^{k+1}}.$$

Take the standard binomial expansion

$$(1 + z)^{-m} = \sum_{n=0}^{\infty} (-1)^n \binom{m+n-1}{n} z^n$$

of the denominator so that

$$\begin{aligned} (R_S + (x - c_B))^{-(k+1)} &= R_S^{-(k+1)} \left(1 + \frac{x - c_B}{R_S}\right)^{-(k+1)} \\ &= R_S^{-(k+1)} \sum_{n=0}^{\infty} (-1)^n \binom{n+k}{n} \left(\frac{x - c_B}{R_S}\right)^n. \end{aligned}$$

Then

$$\sum_{k=0}^p \frac{M_k^S}{(x - c_S)^{k+1}} = \sum_{k=0}^p M_k^S R_S^{-(k+1)} \sum_{n=0}^{\infty} (-1)^n \binom{n+k}{n} \left(\frac{x - c_B}{R_S}\right)^n. \quad (3.35)$$

Thus the multipole expansions of every $S \in \mathcal{I}(B)$ can be re-centred about c_B . For definiteness, we reiterate that the aim here is to form local expansions $L^B(x)$ for each box B , intended to describe the aggregated far-field influence for any particle $x \in B$. Indeed, local expansions take the form

$$L^B(x) = \sum_{n=0}^p L_n^B (x - c_B)^n$$

for coefficients L_n^B , where p is the same parameter as before. We achieve this form by

truncating the n -series in (3.35) to the first p terms. Doing so for a single $S \in \mathcal{I}(B)$ achieves

$$\begin{aligned} \sum_{k=0}^p \frac{M_k^S}{(x - c_S)^{k+1}} &= \sum_{n=0}^p \left((-1)^n \sum_{k=0}^p \binom{n+k}{n} \frac{M_k^S}{R_S^{n+k+1}} \right) (x - c_B)^n \\ &= \sum_{n=0}^p L_{n,S}^B (x - c_B)^n, \quad L_{n,S}^B := (-1)^n \sum_{k=0}^p \binom{n+k}{n} \frac{M_k^S}{R_S^{n+k+1}}, \end{aligned} \quad (3.36)$$

and thus we define the local expansion for a box B as

$$L^B(x) = \sum_{n=0}^p L_n^B (x - c_B)^n, \quad L_n^B = \sum_{S \in \mathcal{I}(B)} L_{n,S}^B. \quad (3.37)$$

At this point every box B at every level of the tree has its own local expansion $L^B(x)$, capturing the far-field influence of everything in B 's interaction list. We finish by pushing these expansions down the tree through each node's children. To be clear, a child's own local expansion is simply its parent's local expansion, re-centred about the child's centre, together with the new interactions the child picks up through its interaction list. This is the 'local-to-local' (L2L) stage of the algorithm. For a box B with child C , with respective centres c_B and c_C , consider $d := c_C - c_B$ so that

$$(x - c_B)^n = ((x - c_C) + d)^n = \sum_{m=0}^n \binom{n}{m} (x - c_C)^m d^{n-m}.$$

Therefore, being careful with the indices here, the local expansions (3.37) become

$$\begin{aligned} L^B(x) &= \sum_{n=0}^p L_n^B \sum_{m=0}^n \binom{n}{m} (x - c_C)^m d^{n-m} \\ &= \sum_{m=0}^p (x - c_C)^m \sum_{n=m}^p \binom{n}{m} L_n^B d^{n-m} =: \sum_{m=0}^p (x - c_C)^m L_m^C. \end{aligned}$$

The local expansion for the child node then accumulates the contribution of the boxes from its interaction list before being passed down to the next level of the tree. We note that the local expansion for the root is zero because it has no boxes in its interaction list. The same is true for both boxes on the first level. Only from the second level of the tree do the local expansions accumulate, and we communicate again that the definition of the interaction list ensures no box is missed and that no box is counted twice.

Once the local expansions reach the leaves, every particle in every leaf has an approximation for the aggregated far-field influence of distant particles. The Coulomb interaction (3.33) is then approximated as this sum of all far field interactions from the local expansion and a direct calculation of the interaction between close neighbours. That is, for a particle λ_k belonging to a leaf L with adjacent leaves $\mathcal{A}(L)$,

$$\sum_{j \neq k} \frac{1}{\lambda_k - \lambda_j} \approx \underbrace{\sum_{k=0}^p L_k^L (\lambda_k - c_L)^k}_{\text{far-field, approx.}} + \underbrace{\sum_{\substack{j \in (L \cup \mathcal{A}(L)) \\ j \neq k}} \frac{1}{\lambda_k - \lambda_j}}_{\text{near, exact.}}. \quad (3.38)$$

For the benefit of the reader, we note that one can draw a correspondence between these steps and the seven steps described in [42] as follows: multipole expansions for the leaves are formed in step one, and then propagated up the tree through their parents in step

two. Step three performs both the M2L and L2L operations together for every level of the tree but the leaves. Step four then performs M2L only for the leaves. Steps five, six, and seven calculate the local expansions and exact interactions for each particle in every leaf, producing the final approximation.

A formal error and complexity analysis of fast multipole methods is given by the original authors [42]. This unfortunately lies outside the scope of this project, but we comment that while the algorithm is of asymptotic order $\mathcal{O}(N)$, it demands substantial computational overhead needed. Several numerical experiments and a discussion of the practical use of fast multipole methods is given in Section 4.6.

3.6 Stochastic Trace Estimation

In Section 3.3 we saw the ratio of determinants (3.23) appear in the transition kernel for the implicit Langevin algorithm. In practice, we work with the log-acceptance probability $\log(\alpha(x, y))$ and are thus interested in the difference

$$\log(|\det(I_N + \Delta t \nabla^2 H_N(x))|) - \log(|\det(I_N + \Delta t \nabla^2 H_N(y))|). \quad (3.39)$$

Computing the determinant of a matrix is not an easy computational task, and in this section we discuss two approaches to do so.

Perhaps the most standard method for calculating (3.39) is based on the Cholesky factorisation. Define the $N \times N$ matrix $A := I_N + \Delta t \nabla^2 H_N(x)$ and consider $A = LL^*$, where L is a lower triangular matrix with real and positive diagonal entries (the Cholesky factorisation of A). Then

$$\det(A) = \det(LL^*) = \det(L)^2,$$

and, since the determinant of a triangular matrix is the product of its diagonal entries,

$$\log(\det(A)) = \log(\det(L)^2) = 2 \log\left(\prod_{i=1}^N L_{ii}\right) = 2 \sum_{i=1}^N \log(L_{ii}). \quad (3.40)$$

The total cost of (3.40) is of order $\mathcal{O}(N^3)$, dominated by the cubic complexity of the Cholesky factorisation [91]. This is unattractive scaling, particularly because the determinant is needed twice at each step of the sampling algorithm, but is certainly simple and effective for moderate values of N .

The second approach we discuss for (3.39) is ‘stochastic Lanczos quadrature’ (SLQ, [93]). An important piece here is the identity

$$\log(\det(A)) = \text{tr}(\log(A)), \quad (3.41)$$

holding for any symmetric positive definite matrix A , where $\text{tr}(\cdot)$ indicates the trace and the matrix logarithm is defined as the inverse to the matrix exponential

$$e^A := \sum_{k=0}^{\infty} \frac{1}{k!} A^k,$$

i.e. $\log(A) := B$ is the matrix satisfying $e^B = A$. To see (3.41), let A be an $N \times N$ symmetric positive definite matrix admitting the spectral decomposition $A = Q\Lambda Q^T$. To be clear, Q is an orthogonal matrix containing the eigenvectors of A , and Λ is a diagonal matrix containing the eigenvalues of A . First,

$$\det(A) = \prod_{i=1}^N \lambda_i \implies \log(\det(A)) = \sum_{i=1}^N \log(\lambda_i),$$

well defined because all eigenvalues are positive. On the other hand, because Q is orthogonal we have $(Q\Lambda Q^T)^k = Q\Lambda^k Q^T$ for all $k \geq 0$, and thus

$$\log(A) = \log(Q\Lambda Q^T) = Q \log(\Lambda) Q^T.$$

Moreover, Λ is diagonal, so $\log(\Lambda) = \text{diag}(\log(\lambda_i))_{1 \leq i \leq N}$. Thus, with the cyclic property of the trace,

$$\text{tr}(\log(A)) = \text{tr}(Q \log(\Lambda) Q^T) = \text{tr}(\log(\Lambda)) = \sum_{i=1}^N \log(\lambda_i).$$

Putting both arguments together gives the claimed identity (3.41). This serves as a bridge to connect SLQ and the log-determinant, but in full generality the algorithm can approximate $\text{tr}(f(A))$ for any A symmetric positive definite and f analytic inside a closed interval [93]. A comprehensive analysis of the algorithm is beyond the scope of this work, but for completeness we provide an outline of the algorithm below.

There are three main steps to SLQ that work together to avoid computing the $\mathcal{O}(N^3)$ eigendecomposition of A . First is the use of a stochastic estimator for the trace. By generating M random vectors $(u_\ell)_{1 \leq \ell \leq M}$ with Rademacher distribution (entries of ± 1 with equal probability) and forming unit vectors $v_\ell = u_\ell / \|u_\ell\|_2$, one has an approximation [93]

$$\text{tr}(f(A)) \approx \frac{N}{M} \sum_{\ell=1}^M v_\ell^T f(A) v_\ell. \quad (3.42)$$

Stochastic trace estimation is often called the ‘Hutchinson trick’ from the original work of Hutchinson [50]. The next step in SLQ is to transform the quadratic form $v^T f(A) v$ because evaluating the matrix function $f(A)$ remains of complexity $\mathcal{O}(N^3)$ for typical f , including the logarithm [48]. Once more consider $A = Q\Lambda Q^T$ and let $\mu := Q^T v$ so that

$$v^T f(A) v = v^T Q f(\Lambda) Q^T v = \mu^T f(\Lambda) \mu = \sum_{i=1}^N f(\lambda_i) \mu_i^2. \quad (3.43)$$

Then, by defining the piecewise-constant function

$$\nu(t) = \begin{cases} 0, & t < \lambda_1, \\ \sum_{j=1}^{i-1} \mu_j^2, & \lambda_{i-1} \leq t < \lambda_i, \quad i = 2, \dots, N, \\ \sum_{j=1}^N \mu_j^2, & \lambda_N \leq t, \end{cases}$$

the sum (3.43) can be seen as the Riemann-Stieltjes integral (see e.g. [77, Chapter 6]) between $a := \lambda_1$ and $b := \lambda_N$,

$$\sum_{i=1}^N f(\lambda_i) \mu_i^2 = \int_a^b f(t) d\nu(t).$$

This integral is then approximated using an $(m+1)$ -point Gauss quadrature rule

$$\int_a^b f(t) d\nu(t) \approx \sum_{k=0}^m \omega_k f(\theta_k),$$

where $\{\omega_k\}$ are the weights and $\{\theta_k\}$ the nodes of the quadrature. These authors use the Lanczos algorithm [38, Chapter 10] to find these weights and nodes. Informally speaking, one starts with the unit vector $w_0 = v_\ell$ and generates an orthonormal basis W_{m+1} for the

Krylov subspace $\text{span}\{w_0, Aw_0, \dots, A^m w_0\}$ so that

$$W_{m+1}^T A W_{m+1} = T_{m+1},$$

where T_{m+1} is an $(m+1) \times (m+1)$ tridiagonal matrix. Both $\{\omega_k\}$ and $\{\theta_k\}$ can be obtained from the eigendecomposition of T_{m+1} , and since it is tridiagonal, this is only of order $\mathcal{O}(m^2)$ [91]. Each step of the Lanczos algorithm requires a matrix-vector product, so the total cost of finding the quadrature information is therefore $\mathcal{O}(mN^2 + m^2)$. In sum,

$$\text{tr}(f(A)) \approx \frac{N}{M} \sum_{\ell=1}^M v_\ell^T f(A) v_\ell = \frac{N}{M} \sum_{\ell=1}^M \sum_{i=1}^N f(\lambda_i) \mu_{\ell,i}^2 \approx \frac{N}{M} \sum_{\ell=1}^M \sum_{k=0}^m \omega_{\ell,k} f(\theta_{\ell,k}),$$

and the total cost of the algorithm is approximately $\mathcal{O}(Mm(N^2 + m))$. Typical values for both m and M are less than 100 in the experiments presented by the original authors [93], even for systems with dimension as large as $N = 10^6$. We present our own numerical experiments for the stochastic Lanczos quadrature in Section 4.6.

3.7 Random Matrix Alternatives

This chapter has focused solely on the numerical simulation of Dyson Brownian motion (3.1), motivated by its utility as a sampling scheme for random matrices. However, as briefly discussed in the introduction, Dyson Brownian motion is certainly not the only approach one can take for this sampling problem. In this section we explore other models that have been proposed in the literature, namely [25, 69, 36]. Summary tables for all algorithms, including those that use Dyson Brownian motion, can be found in Section 3.8.

To begin we discuss the tridiagonal random matrix model from Dumitriu and Edelman [25]. It applies to only the β -Hermite, β -Laguerre, and β -Jacobi ensembles, the first two of which we have seen previously (Sections 2.1 and 2.6) and thus focus on here. Importantly, it provides an exact sampling algorithm for these ensembles for any $\beta > 0$. The scheme is easily understood; they show that the $N \times N$ tridiagonal matrix

$$H_\beta \sim \frac{1}{\sqrt{2}} \begin{pmatrix} \mathcal{N}(0,2) & \chi_{(N-1)\beta} & & & \\ \chi_{(N-1)\beta} & \mathcal{N}(0,2) & \chi_{(N-2)\beta} & & \\ & \ddots & \ddots & \ddots & \\ & & \chi_{2\beta} & \mathcal{N}(0,2) & \chi_\beta \\ & & & \chi_\beta & \mathcal{N}(0,2) \end{pmatrix} \quad (3.44)$$

has eigenvalues that precisely coincide with the eigenvalues of the β -Hermite ensembles, i.e. the β -ensembles (2.6) for the quadratic potential case, $V(x) = x^2/2$. Furthermore, if X_β is an $N \times M$ matrix and $a := \beta M/2$, then the $N \times N$ matrix L_β

$$L_\beta = X_\beta X_\beta^T, \quad X_\beta \sim \begin{pmatrix} \chi_{2a} & & & & \\ \chi_{\beta(N-1)} & \chi_{2a-\beta} & & & \\ & \ddots & \ddots & & \\ & & \chi_\beta & \chi_{2a-\beta(N-1)} & \end{pmatrix} \quad (3.45)$$

has eigenvalues that precisely coincide with the eigenvalues of the Laguerre β -ensembles (2.32). In both (3.44) and (3.45), χ_k refers to the chi distribution, the density of which is

$$f(x; k) = \frac{1}{2^{k/2-1} \Gamma(\frac{k}{2})} x^{k-1} e^{-x^2/2}$$

for all $x \geq 0$. We note that both constructions correspond to the choice of variance $\sigma^2 = 1$, meaning that the eigenvalues of H_β concentrate on $[-\sqrt{2\beta N}, \sqrt{2\beta N}]$. Scaling

$H_\beta \mapsto \sqrt{2/(\beta N)} H_\beta$ achieves the typical support $[-2, 2]$, and scaling $L_\beta \mapsto 1/(\beta M) L_\beta$ achieves the typical support $[(1 - \sqrt{c})^2, (1 + \sqrt{c})^2]$ for $c := N/M$ [25].

The tridiagonal random matrix model is very popular because it provides an exact sampling scheme of $\mathcal{O}(N^2)$ complexity. Indeed, constructing the matrix is $\mathcal{O}(N)$, and finding the eigenvalues of a symmetric tridiagonal matrix is $\mathcal{O}(N^2)$ [91]. Its primary limitation is the fact it is restricted to these three classical ensembles. This is at least one factor that motivated the work of Olver et al. [69]. They provide an exact (up to machine precision) sampling scheme for the invariant unitary ensembles; that is, a sampling scheme for ensembles from (2.6) for any arbitrary potential V and $\beta = 2$. The algorithm exploits the determinantal point process that we have explored in detail in Section 2.3. Indeed, to quote these authors, the algorithm comprises two main steps.

1. Construct the orthogonal polynomials associated with the potential V .
2. Sample a determinantal point process defined through this sequence of orthogonal polynomials.

The complexity of the algorithm is dominated by the sampling of the determinantal point process and is of asymptotic $\mathcal{O}(N^3)$ complexity.

Finally, let us discuss the algorithm proposed more recently by Gautier et al. [36]. It can be seen as an extension of the tridiagonal random matrix model [25] and relies on the reparametrisation of the β -ensembles (2.6) as the eigenvalues of a random Jacobi matrix. Indeed, consider the $N \times N$ symmetric tridiagonal matrix

$$J_{a,b} = \begin{pmatrix} a_1 & \sqrt{b_1} & & & \\ \sqrt{b_1} & a_2 & \sqrt{b_2} & & \\ & \ddots & \ddots & \ddots & \\ & & \sqrt{b_{N-2}} & a_{N-1} & \sqrt{b_{N-1}} \\ & & & \sqrt{b_{N-1}} & a_N \end{pmatrix}, \quad (3.46)$$

where a and b are jointly distributed according to

$$f(a, b) \propto \exp\left(-\frac{\beta}{2} \text{tr}(V(J_{a,b}))\right) \prod_{k=1}^{N-1} b_k^{\frac{\beta}{2}(N-k)-1}. \quad (3.47)$$

Then the eigenvalues of $J_{a,b}$ follow the same joint probability density as that of the β -ensembles (2.6) [36]. Notice that (3.47) eliminates the Vandermonde determinant, a large source of difficulty for schemes targeting Dyson Brownian motion. Gautier et al. employ a Gibbs sampler for (3.46) and assert that the chain mixes in time $\mathcal{O}(\log N)$, meaning the complexity of the algorithm is dominated by the $\mathcal{O}(N^2)$ eigendecomposition of $J_{a,b}$. Unlike the tridiagonal model of Dumitriu and Edelman, this is an approximate sampling scheme, but it extends to arbitrary convex polynomial potentials.

3.8 Algorithm Summaries

We conclude this chapter by providing two summary tables for the algorithms discussed. Table 1 contains the sampling schemes that use Dyson Brownian motion and Table 2 contains the alternative random-matrix or determinantal point process sampling schemes. To accompany the tables we also provide brief discussion of DBM-based schemes below.

1. **Unadjusted methods.** We have discussed three variants of Euler methods and an implicit midpoint method. See Sections 3.1, 3.2, and 3.3.
 - (a) Explicit Euler (ULA). This is the standard Euler-Maruyama discretisation of (3.1). We can expect to need a minimum scaling of $\Delta t \sim \mathcal{O}(N^{-5/3})$ in order to prevent eigenvalues from crossing.

- (b) Tamed Euler. Proposed by Li and Menon [57], this scheme tames the potential V but leaves the Coulomb interaction unchanged. It has the same problem with eigenvalue crossings.
- (c) Implicit Euler (ILA). This is the implicit (backwards) Euler discretisation of the dynamics. It solves a minimisation problem at each step and is accordingly more expensive, but it prevents any eigenvalue crossing from occurring.
- (d) Implicit Midpoint (IMLA). Proposed by [56] for Bayesian imaging problems. It too needs to solve a minimisation problem at each step but does not prevent eigenvalues from crossing.

2. **Adjusted methods.** The unadjusted methods can be used to produce proposals in a Metropolis-Hastings framework. We have also discussed a one-step Hamiltonian Monte Carlo method. If the proposal has any two eigenvalues cross it will always be rejected. See Sections 3.2, 3.3, and 3.4.

- (a) Metropolis-Adjusted Langevin Algorithm (MALA). This uses explicit Euler as a proposal. Thus we can expect the same minimum scaling $\Delta t \sim \mathcal{O}(N^{-5/3})$.
- (b) One-step Hamiltonian Monte Carlo (HMC). Proposed by Chafaï and Ferré for simulating log-gases, including the case of (3.1). With only one step the proposals can be seen as equivalent to a MALA proposal.
- (c) Metropolis-Adjusted Implicit Midpoint Langevin Algorithm (MAIMLA). This uses the implicit midpoint as a proposal. One reason it is desirable is that the Jacobian in the transition kernel is symmetric, so no determinant needs to be calculated. The proposal may still have two eigenvalues cross.
- (d) Metropolis-Adjusted Implicit Langevin Algorithm (MAILA). This uses implicit Euler as a proposal, so rejections are never the result of a crossing. However, two determinants must be calculated for the Jacobian in the transition kernel.

Table 1: A summary of the Dyson Brownian motion-based sampling schemes.

Name	Reference	Bias	Crossings	Per-Step Cost
ULA	Section 3.1	✓	✓	$\mathcal{O}(N^2)$ or $\mathcal{O}(N)$ with FMM.
Tamed	Li & Menon [57]	✓	×	$\mathcal{O}(N^2)$ or $\mathcal{O}(N)$ with FMM.
ILA	Section 3.1	✓	✓	$\mathcal{O}(N^2)$ (Newton, CG).
IMLA	Klatzer et al. [56]	✓	✓	$\mathcal{O}(N^2)$ (Newton, CG).
MALA	Section 3.3	×	×	$\mathcal{O}(N^2)$ or $\mathcal{O}(N)$ with FMM.
HMC	Chafaï & Ferré [18]	×	×	$\mathcal{O}(N^2)$ or $\mathcal{O}(N)$ with FMM.
MAIMLA	Section 3.3	×	×	$\mathcal{O}(N^2)$ (Newton, CG).
MAILA	Section 3.3	×	×	$\mathcal{O}(N^3)$ or $\mathcal{O}(N^2)$ with SLQ.

Table 2: A summary of the alternative sampling schemes.

Reference	Principle	Beta	Potential	Notes
Dumitriu & Edelman [25]	Tridiagonal Random Matrix	$\beta > 0$	Classical (Hermite, Laguerre, Jacobi)	Exact, $\mathcal{O}(N^2)$.
Olver et al. [69]	Determinantal Point Process	$\beta = 2$	Sufficiently Smooth	Exact, $\mathcal{O}(N^3)$.
Gautier et al. [36]	Gibbs Sampler on Tridiagonal	$\beta > 0$	Convex Polynomials	Approx, $\mathcal{O}(N^2)$.

4 Numerical Experiments

This chapter serves to present a range of numerical experiments for the algorithms discussed in Chapter 3. Unless explicitly stated, the code for these experiments is our own and is written in Python.

4.1 Performance Metrics

We primarily use three metrics to evaluate the performance of each sampling scheme. The first of these is a qualitative visual comparison of a histogram of the obtained samples against the limiting distribution, if known. Of course, the number of bins in the histogram influences the resolution of this comparison and we will state the number used. For a more formal evaluation we also compute the Kolmogorov-Smirnov (KS) and p -Wasserstein distances from the cumulative distribution function (cdf) of the obtained samples to the limiting distribution. Indeed, we use

$$F_{n,M}(\lambda) = \frac{1}{NM} \sum_{j=1}^M \sum_{k=1}^N \mathbb{I}(\lambda_k^{n,j} \leq \lambda) \quad (4.1)$$

to denote the empirical spectral distribution of an N -dimensional system with M independent realisations. The notation $\lambda_k^{n,j}$ indicates it is recorded at time step n of trial j . Let us focus, for now, on the β -ensembles (2.6). In this case, let

$$F_N(\lambda) = \frac{1}{N} \int_{-\infty}^{\lambda} K_N(x, x) dx \quad (4.2)$$

be the finite cdf obtained from the kernel as described in Section 2.3. Then the KS distance is defined as the supremum,

$$D_{n,M}^{\text{KS}} := \sup_{\lambda} |F_{n,M}(\lambda) - F_N(\lambda)|, \quad (4.3)$$

and the p -Wasserstein distance is defined

$$D_{n,M}^{\text{W}} = \left(\int_0^1 |F_{n,M}^{-1}(\lambda) - F_N^{-1}(\lambda)|^p d\lambda \right)^{1/p}. \quad (4.4)$$

In practice, we prefer to work with the cdf of the limiting distribution instead of F_N when it is known. This is because (a) the finite distribution converges to the limiting distribution very quickly (see e.g. [57] and later discussion), and (b) great care is needed to accurately compute K_N , even for moderate values of N [10]. For these reasons, experiments concerning the β -ensembles will focus on the following three potentials.

1. The quadratic case $V(x) = x^2/2$, for which the semicircle law

$$\rho(x) = \frac{1}{2\pi} \sqrt{4 - x^2} \mathbb{I}(|x| \leq 2)$$

is the limiting distribution [58]. For $|x| \leq 2$, the cdf is given by

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \left(\frac{x\sqrt{4 - x^2}}{4} + \arcsin\left(\frac{x}{2}\right) \right).$$

2. The mixed quartic case $V(x) = qx^2/2 + gx^4/4$, for which the law

$$\rho(x) = \frac{1}{2\pi}(1 + 2ga^2 + gx^2)\sqrt{4a^2 - x^2}\mathbb{I}(|x| \leq 2a), \quad a = \sqrt{\frac{\sqrt{1+12g} - 1}{6g}}$$

is the limiting distribution when $q = 1$ [8]. For $|x| \leq 2a$, the cdf is given by

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \left[\frac{x(2 + 2ga^2 + gx^2)}{8} \sqrt{4a^2 - x^2} + \arcsin\left(\frac{x}{2a}\right) \right].$$

3. The ‘Claeys’³ potential $V(x) = x^4/20 - 4x^3/15 + x^2/5 + 8x/5$, for which the law

$$\rho(x) = \frac{1}{10\pi}(x+2)^{1/2}(2-x)^{5/2}\mathbb{I}(|x| \leq 2)$$

is the limiting distribution [19]. For $|x| \leq 2$, the cdf is given by

$$F(x) = \frac{1}{\pi} \left[\frac{(3x^3 - 16x^2 + 18x + 64)\sqrt{4-x^2}}{120} + 2\arcsin\left(\frac{\sqrt{x+2}}{2}\right) \right].$$

One may recall that the limiting densities for these cases were illustrated in Figure 3. We have also seen that the limiting distribution of the Wishart-Laguerre ensembles is the Marčenko-Pastur law (Figure 7)

$$\frac{1}{2\pi c\lambda} \sqrt{(\lambda - a)(b - \lambda)} \mathbb{I}(a \leq \lambda \leq b),$$

where $a := (1 - \sqrt{c})^2$ and $b := (1 + \sqrt{c})^2$. For $c \leq 1$ and $a \leq x \leq b$, the cdf takes the closed form [61]

$$F(x) = 1 + \frac{1}{2\pi c} \left[\sqrt{(x-a)(b-x)} - (1+c) \arccos\left(\frac{2x-a-b}{b-a}\right) + 2(1-c) \arctan\left(\sqrt{\frac{a(b-x)}{b(x-a)}}\right) \right].$$

Eigenvalues of the complex Ginibre ensemble are known to converge to the circular law (Figure 6). For finite dimensions one may use instead use the distribution [61]

$$\rho_N(z) = \frac{1}{\pi} e^{-N|z|^2} \sum_{k=0}^{N-1} \frac{|\sqrt{N}z|^{2k}}{k!},$$

and it is common to express this in terms of the radius $r := |z|$, for which the limiting density becomes $\rho(r) = 2r$ for $0 \leq r \leq 1$ [61]. The limiting cumulative distribution function is therefore $F(r) = r^2$ for $0 \leq r \leq 1$.

Let us now address the convergence of the finite distribution (4.2) to the limiting laws. First is the matter of numerically calculating the kernel K_N . We do this by building a family of polynomials $\{\pi_k\}$ orthogonal with respect to $e^{-NV(x)}$ through the general, potential-dependent case of the recurrence relation (2.20) (see [86, 37], for instance)

$$\pi_{k+1}(x) = (x - a_k)\pi_k(x) - b_k\pi_{k-1}(x), \quad \pi_{-1} \equiv 0, \quad \pi_0 \equiv 1. \quad (4.5)$$

For $k \geq 1$ the recurrence coefficients are calculated over a fine grid with spacing Δx ,

$$a_k = \frac{\langle x\pi_k, \pi_k \rangle}{\langle \pi_k, \pi_k \rangle}, \quad b_k = \frac{\langle \pi_k, \pi_k \rangle}{\langle \pi_{k-1}, \pi_{k-1} \rangle}, \quad \langle f, g \rangle := \sum_j f(x_j)g(x_j)e^{-NV(x_j)}\Delta x.$$

³Recall that we have chosen to name this potential for the lead author [19].

Calculating the coefficients in this way is sometimes known as the discretised Stieltjes procedure and is far more stable than an application of the Gram-Schmidt process [37]. After finding the normalising constants $c_k^2 = \langle \pi_k, \pi_k \rangle$ we form the orthonormal wavefunctions

$$\psi_k(x) = \frac{1}{c_k} e^{-NV(x)/2} \pi_k(x), \quad k = 0, \dots, N-1,$$

and then assemble the kernel

$$K_N(x_i, x_j) = \sum_{k=0}^{N-1} \psi_k(x_i) \psi_k(x_j).$$

We note that one could alternatively use the Christoffel-Darboux formula (2.21) for this step. Finally, the cdf (4.2) is obtained through standard numerical integration.

The top panels of Figure 9 provide a comparison of the finite ($N = 30$) and limiting distribution functions of the β -ensembles for each of the three aforementioned potentials (the quadratic case $V(x) = x^2/2$, the quartic case $V(x) = x^4/4$, and the Claeys case $V(x) = x^4/20 - 4x^3/15 + x^2/5 + 8x/5$). Notice that even for $N = 30$ there is a strong agreement between the two. Indeed, for the $N = 30$ quadratic case we find the KS distance between $F_N(\lambda)$ and the limiting distribution $F_\infty(\lambda)$ to be approx. 0.00103, in agreement with the authors [57]. In the bottom panel we produce a log-log plot of the distances between the finite and limiting densities for each potential over a range of $N \in [2, 50]$. The distance appears to scale like $\mathcal{O}(N^{-1})$ which too is in agreement with theoretical results [2, 39].

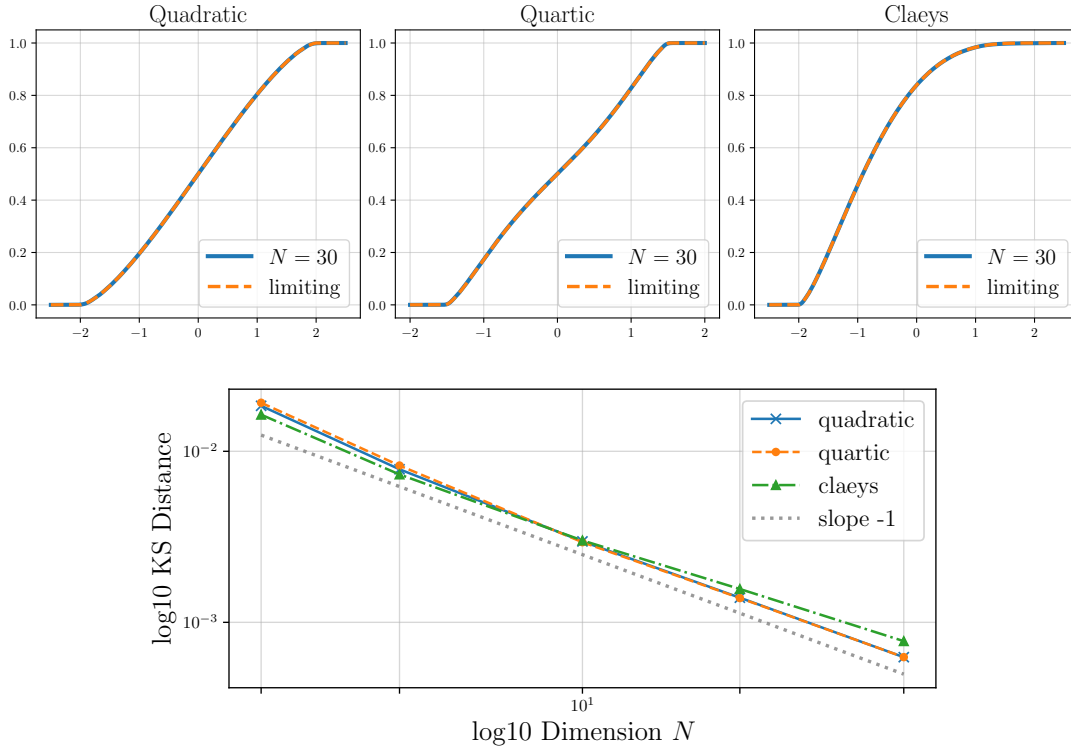


Figure 9: Top: the finite and limiting cumulative distribution functions for three cases of the β -ensembles (quadratic $V(x) = x^2/2$, quartic $V(x) = x^4/4$, and Claeys $V(x) = x^4/20 - 4x^3/15 + x^2/5 + 8x/5$). Bottom: a log-log plot of the Kolmogorov-Smirnov distance between the finite and limiting distributions for $N \in \{2, 4, 10, 22, 50\}$. A line of slope -1 is given for comparison.

4.2 Explicit Schemes and Crossings

Analysis in Section 3.2 suggested that in order for the explicit Euler scheme to produce trajectories with no collisions, the step size must scale like $\mathcal{O}(N^{-5/3})$. To test this conjecture we consider the following experiment.

1. Fix a set of dimensions $\{N\}$. For each N , define a grid of steps sizes $\{\Delta t\}$ finely spaced over a chosen interval, and draw M independent realisations of eigenvalues from the GUE (e.g. by explicit construction or from the tridiagonal random matrix model [25]).
2. For each $(N, \Delta t)$ pair, apply one explicit Euler step to each of the M realisations. Record the fraction of realisations in which any two eigenvalues cross.
3. For each N , report the largest Δt for which the crossing fraction, and that of every smaller Δt in the grid, is zero.

We perform this experiment with a set of five dimensions N log-spaced over $[50, 200]$, a grid of 200 step sizes log-spaced over $[N^{-2}, N^{-1}]$, and set $M = 10^4$. This procedure is then repeated for the tamed Euler and implicit midpoint schemes for comparison. The results, illustrated as a log-log plot of the largest step size and dimension, can be seen in Figure 10. They are noisy, likely because of the small number of realisations and discrete grid of step sizes. Moreover, it is important to clarify that the previous analysis depended on the minimum gap for the GUE scaling like $\delta \sim \mathcal{O}(N^{-4/3})$ and that this is obtained in the limit $N \rightarrow \infty$, so in practice a larger range of N should be tested. Nonetheless, a comparison of the line $y = N^{-5/3}$ does not suggest that this scaling is an unreasonable estimate. It is not surprising to see that the tamed Euler scheme exhibits almost the same behaviour as the explicit scheme because they both treat the repulsion interaction in the same way. Similarly, it is not too surprising to see IMLA yield a comparatively much larger step size, because the midpoint step dampens the effective drift towards a colliding pair.

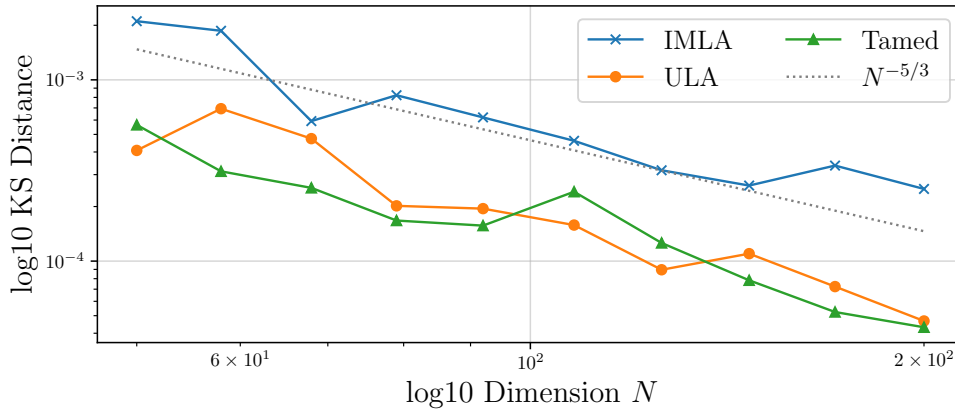


Figure 10: The largest step size needed for one step of the explicit Euler (ULA), tamed Euler, and implicit midpoint (IMLA) schemes to produce no crossings over $M = 10^4$ independent realisations. The line $y = N^{-5/3}$ is provided for comparison.

4.3 Unadjusted Implicit Schemes

A major advantage of the implicit Euler scheme (3.5) is that, in theory, it prevents any two eigenvalues from crossing irrespective of the step size Δt . In fact, the method is unconditionally stable in the sense that we can choose $\Delta t = \mathcal{O}(1)$. This is very important: the explicit schemes discussed in Section 4.2 demand strict scaling on the step size because of the challenges they face regarding stability and crossings. The purpose of this section is to explore the consequences of choosing a large step size.

We begin with a preliminary investigation into the macroscopic convergence of the dynamics (1.2) when $V(x) = x^2/2$, $\beta = 1$, and $\Delta t = 0.1$. We consider $N = 200$ particles across $M = 100$ independent trials initialised uniformly on the interval $[-2, 2]$. Snapshots of the system are taken every step in the second half of the trajectory for $T = 20.0$. We find that the empirical histograms agree with the semicircle law very well. So, we increase N to 500 and repeat the same experiment, then once more with $N = 1000$. For a quantitative metric we also calculate the KS distance between the snapshots and the limiting distribution for all three cases. An illustration of these results can be found in Figure 11. Despite the $\mathcal{O}(1)$ step size, convergence to the limiting distribution seems to only improve as N increases. This is a striking result and motivates exploration of other aspects of the algorithm. We find that the number of Newton iterations is consistent irrespective of dimension; after rounding to the nearest integer, the average is 7 in each case. The number of CG iterations does increase, with the average number being 28, 40, and 50 per Newton iteration, although this is to be expected as the algorithm depends on the size of the linear system and neither result is of concern.

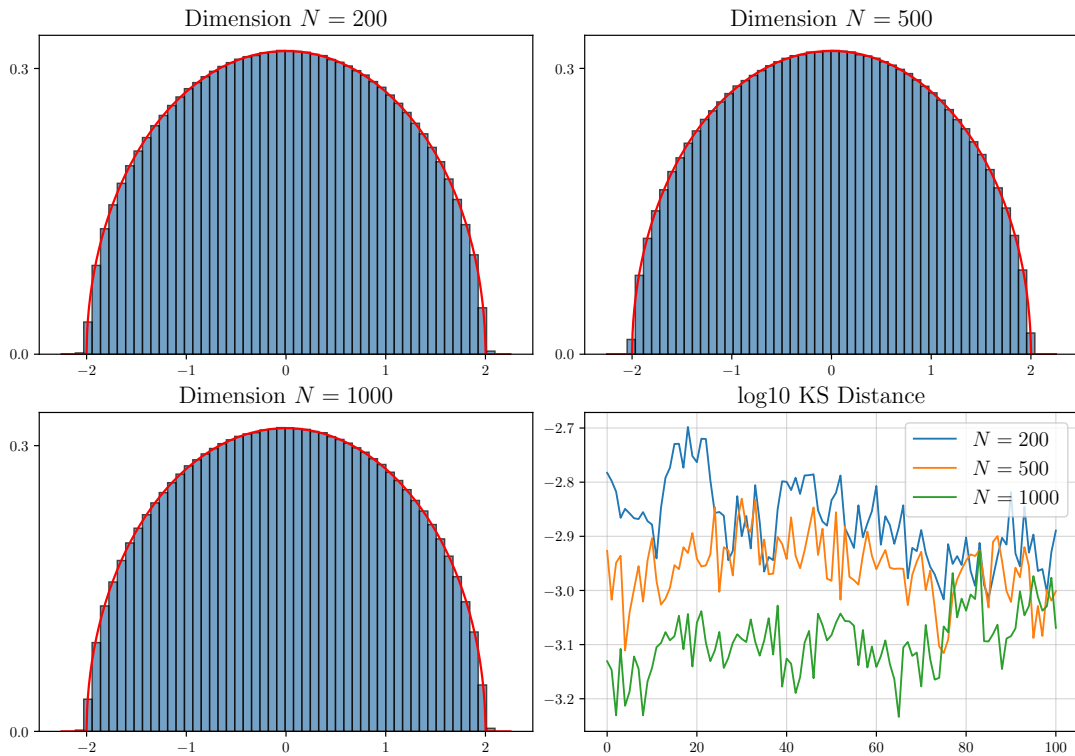


Figure 11: Macroscopic performance of the implicit Euler scheme applied to a system with $V(x) = x^2/2$, $\beta = 1$, and $\Delta t = 0.1$. The limiting distribution is the semicircle law and is given in red, and each of these histograms use 80 bins.

We now pivot from analysis of the macroscopic behaviour and focus on the microscopic behaviour. For each snapshot we consider the middle 80% of particles to avoid the inconsistent behaviour near the edges, and apply the scaling (2.22) to achieve a set of spacings normalised to have mean one. We collect all snapshots and plot these as a histogram, together with Wigner’s surmise (2.23) for $\beta = 1$, in Figure 12a. It is clear that on this level the scheme faces major challenges. Indeed, the eigenvalues are far from resembling the theoretical spacing distribution, with behaviour that deteriorates with N . We suggest this is almost certainly a consequence of the bias introduced by large step size $\Delta t = 0.1$. To this end, we repeat the same simulation for $N = 200$ but now vary $\Delta t \in \{0.01, 0.05, 0.001\}$. See the results in Figure 12b. In this regime the spacing distributions match the theoretical curve far better as Δt decreases.

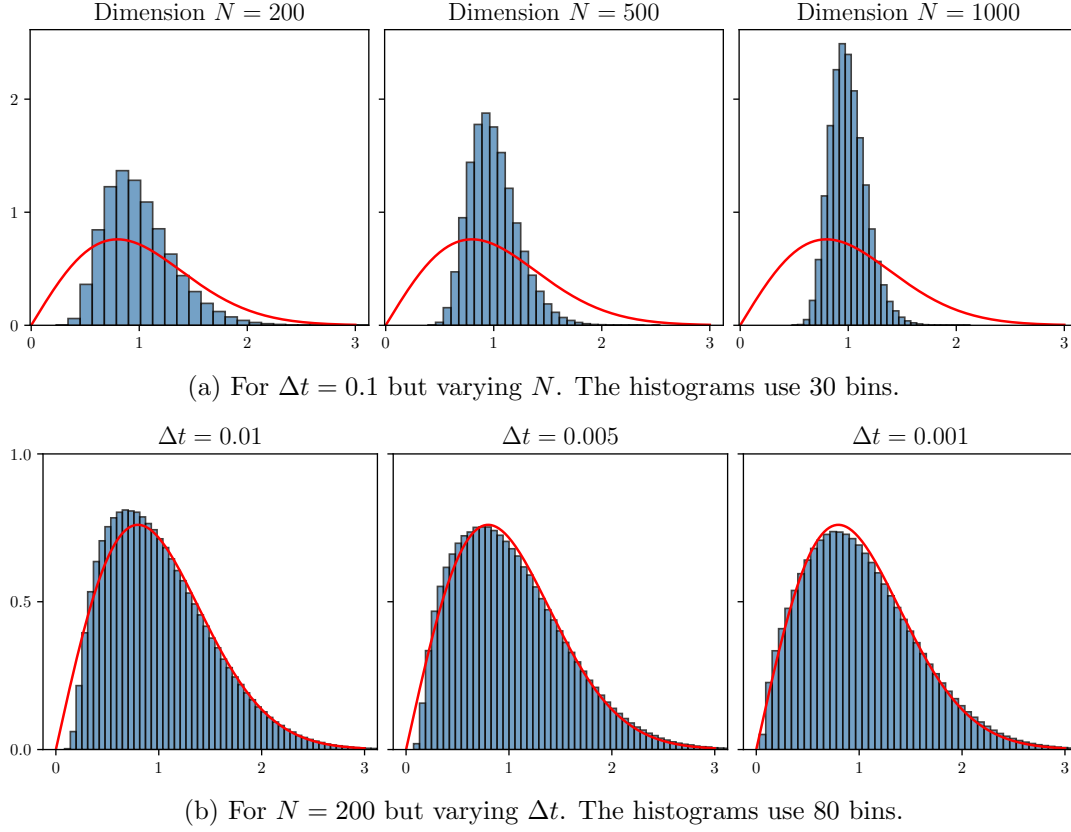


Figure 12: Microscopic performance of the implicit Euler scheme. Superimposed in red is Wigner's surmise for $\beta = 1$.

It is very interesting to us that the bias introduced by the implicit scheme does not seem to materialise on the macroscopic scale. We suggest that it may be a consequence of the mean-field scaling. That is, that the contribution

$$\frac{1}{N} \sum_{j \neq k} \frac{1}{\lambda_k - \lambda_j}$$

is $\mathcal{O}(1)$ as $N \rightarrow \infty$. In this case the scale of the dynamics (3.1) is $\mathcal{O}(1)$ and, as $N \rightarrow \infty$, the empirical density

$$\mu_N := \frac{1}{N} \sum_{j=1}^N \delta_{\lambda_j}$$

has improved empirical convergence to the limiting distribution. A possible approach to show this could, for instance, follow the reasoning in Section 3.2 where one bounds ‘close’ and ‘far’ sets of eigenvalues, using known theory on extreme eigenvalue spacings to bound the contribution from the ‘close’ set above by a constant independent of N . We choose to not pursue this point here, leaving it instead for future work.

4.4 Metropolis-Hastings Algorithms

This section concerns itself with experiments for the Metropolis-Hastings algorithms described in Section 3.3. The first item we address is the one-step Hamiltonian Monte Carlo algorithm discussed in Section 3.4. There we showed that, when only one step is taken, proposals from this algorithm can be seen as equivalent to a MALA proposal, simply with the variance of the dynamics rescaled. Indeed, we showed that despite the input step size scaling like $\Delta t_{\text{HMC}} = \sqrt{2\Delta t_{\text{MALA}}}$, both schemes need the same number of steps to

make the same progress on the dynamics. To illustrate this phenomenon we simulate the dynamics (3.1) with a quartic potential, $V(x) = x^4/4$, but initialise from the GUE (i.e. the semicircle distribution). It suffices to consider a dimension of $N = 50$ across $M = 1000$ independent realisations. Let us set an effective time of $T = 3.0$ with

$$\Delta t_{\text{MALA}} = \frac{1}{N^2}, \quad n_{\text{MALA}} = \frac{3}{\Delta t_{\text{MALA}}} = 7500, \quad \Delta t_{\text{HMC}} = \sqrt{\frac{2}{N^2}}, \quad n_{\text{HMC}} = \frac{3}{\Delta t_{\text{HMC}}} = 106,$$

where n denotes the number of steps taken by the algorithm. We collect only the final positions of all M realisations together to form the histograms in Figure 13a. Notice that MALA appears to have reached the limiting distribution, but the one-step HMC algorithm has hardly progressed from its initialisation. The results of the same experiment, but with $n_{\text{HMC}} = n_{\text{MALA}} = 7500$, can be seen in Figure 13b. There both algorithms appear to reach the limiting distribution. In sum, we find that this experiment agrees with the theoretical analysis: that while the one-step HMC algorithm appears to need a smaller step size, this is illusory and indeed needs the same number of total steps as MALA.

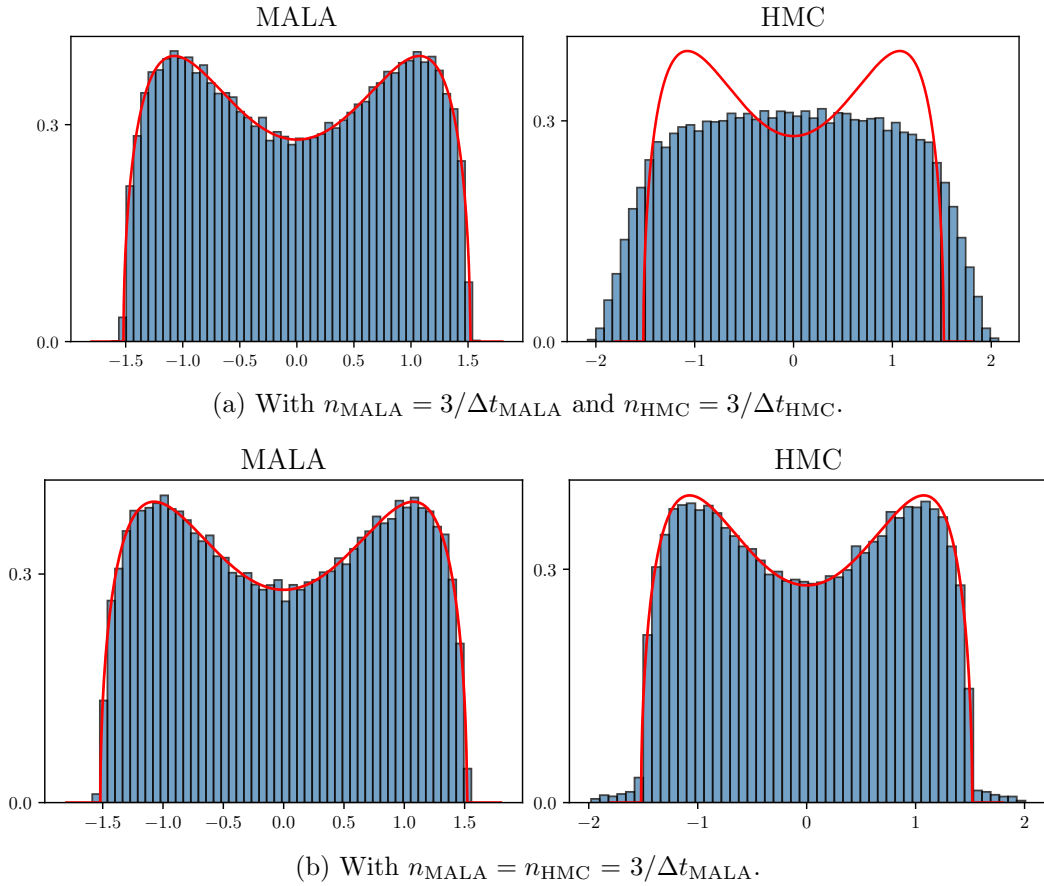


Figure 13: The collective final positions of $M = 1000$ independent trajectories of the dynamics (3.1) with $N = 50$, a quartic potential $V(x) = x^4/4$, and $\beta = 2$ run for an effective time of $T = 3.0$. Trajectories are initialised from the semicircle distribution. Histograms are plotted with 50 bins and the red curve is the limiting distribution. We emphasise that ‘HMC’ refers to the one-step Hamiltonian Monte Carlo algorithm described in Section 3.4.

Now we move to the discussion of a related subject. We ask this: for each of the Metropolis algorithms (MALA, MAIMLA, MAILA, and one-step HMC), what is the necessary step size scaling needed in order to achieve a desired target probability? Such scaling is important because it controls the complexity of the algorithms. To find it, we

run an experiment that adaptively tunes the step size over a large number of steps until it converges. Simply put, if the acceptance rate is too high, the step size is increased, and if the acceptance rate is too low, the step size is decreased. More formally, consider a target acceptance rate α^* and the acceptance rate achieved at step k , denoted α_k . Then we run the update

$$(\Delta t)_{k+1} = (\Delta t)_k + (\Delta t)_k \left(\frac{\alpha_k - \alpha^*}{k^\eta} \right), \quad k = 0, \dots, K \quad (4.6)$$

for a large K , with η chosen so that

$$\sum_{k=1}^{\infty} \frac{1}{k^\eta} = \infty, \quad \sum_{k=1}^{\infty} \frac{1}{k^{1+\eta}} < \infty.$$

We apply this update to a set of ten dimensions N log-spaced between $N = 30$ and $N = 500$. The update (4.6) has the acceptance rate averaged over $M = 100$ independent trials at each iteration k . To ensure K is chosen large enough for convergence, we check the trace plots of both the acceptance rate and step size; refer to Figure 14 for an example of successful convergence. To find the necessary scaling we produce a log-log plot of the final step sizes and dimension. For this, refer to Figure 15.

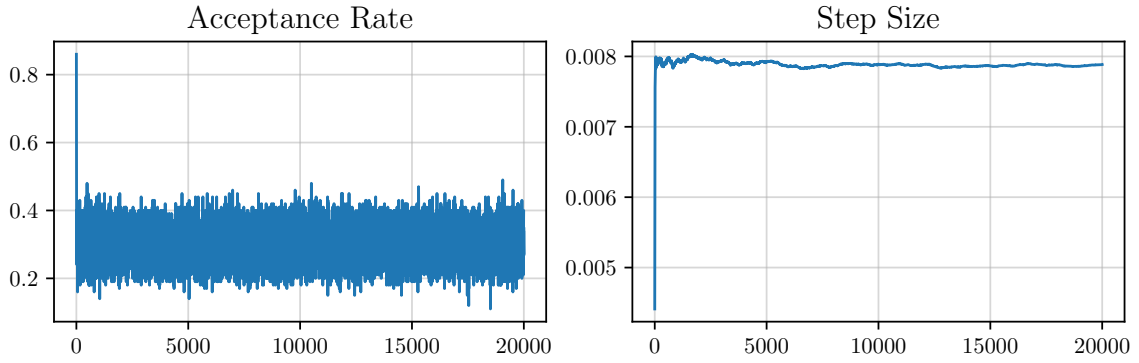


Figure 14: Convergence of the step size under the adaptive scheme (4.6). In this instance the target is 30% and the dynamics are simulated with MALA for $N = 70$.

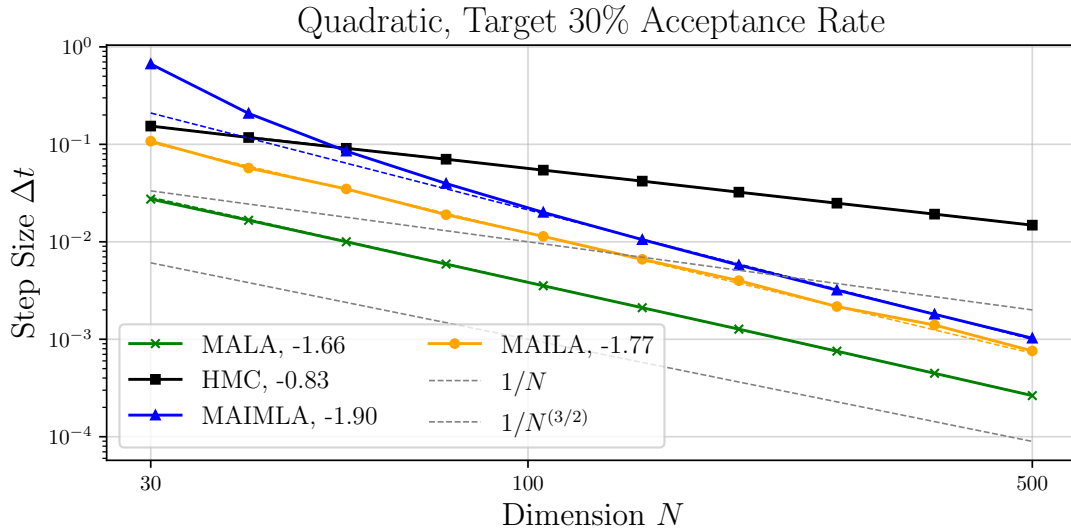


Figure 15: The final step size of the four Metropolis-adjusted methods under the scheme (4.6) targeting a 30% acceptance rate. The system has a quadratic potential and $\beta = 2$, with acceptance rates averaged over $M = 100$ independent trials at each iteration.

There are several important results to unpack here, so let us discuss them one by one. First we comment on the scaling for MALA. It is known that when the target is a product measure one should scale $\Delta t \sim \mathcal{O}(N^{-1/3})$ and achieve the celebrated acceptance rate of approx. 0.574 [75]. Now, our target

$$\pi(x) \propto \exp(-\beta N H_N(x)) = \exp\left(-\beta N \left(\frac{1}{2} \sum_{j=1}^N V(x_j) - \frac{1}{N} \sum_{j < k} \log(x_k - x_j)\right)\right)$$

is clearly not a product measure because of the repulsion interaction, and the observed result of $\Delta t \sim \mathcal{O}(N^{-5/3})$ is almost certainly a consequence of the additional scaling needed to prevent eigenvalues from crossing. Indeed, it agrees with the analysis in Section 3.2 and the experiment described in Section 4.2. On a similar theme, we briefly comment that it is reassuring to see that the slope of the one-step HMC scheme is precisely half of MALA's, as this agrees with then notion that $\Delta t_{\text{HMC}} = \sqrt{2\Delta t_{\text{MALA}}}$.

Now consider the observed scaling for MAILA. It is not immediately clear to us why this is worse than its explicit counterpart, MALA, because the algorithm does not face any issues with eigenvalues crossing. At this time we suggest that despite this apparent strength, the repulsion mechanism is still the probable culprit. Indeed, the Hessian

$$[\nabla^2 H_N(x)]_{k\ell} = \begin{cases} -\frac{1}{N(x_k - x_\ell)^2}, & k \neq \ell, \\ \frac{1}{2} V''(x_k) + \frac{1}{N} \sum_{j \neq k} \frac{1}{(x_k - x_j)^2}, & k = \ell. \end{cases}$$

appears in several places, namely in the Newton solver and in the determinant ratio

$$\frac{|\det(I_N + \Delta t \nabla^2 H_N(x))|}{|\det(I_N + \Delta t \nabla^2 H_N(y))|}.$$

Of particular importance is the highly singular squared gap term

$$\frac{1}{N} \sum_{j \neq k} \frac{1}{(x_k - x_j)^2}.$$

We have previously discussed that one can expect the average eigenvalue spacing to scale like $\mathcal{O}(1/N)$ on average, and in some cases - e.g. for the GUE - theoretical results show that typical extreme gaps scale like $\mathcal{O}(N^{-4/3})$ [7]. The Hessian is therefore of order $\mathcal{O}(N^{5/3})$, and perhaps $\Delta t \sim \mathcal{O}(N^{-5/3})$ is needed to balance this contribution.

Finally, let us consider the observed MAIMLA scaling. This algorithm exhibits some nonlinear behaviour for small dimensions, and we note that the illustrated linear fit omits these points. Now, this algorithm is subject to both the eigenvalue crossing constraint and the conditioning of the Newton solver, it is not unreasonable to expect $\Delta t \sim \mathcal{O}(N^{-5/3})$ appearing as a consequence of either interaction. We suggest that the observed scaling seen in Figure 15 is likely an artefact of (a) the small range of dimensions N , and that the slope would approach $-5/3$ in the limit $N \rightarrow \infty$, and/or (b) the parameters of this particular system, particularly that the target acceptance probability is 0.3. Indeed, when we increase the target acceptance probability to 70% we see the observed slope for the same experiment decrease in magnitude to -1.75 .

To finish our discussion of the Metropolis-adjusted methods we perform some numerical experiments that demonstrate their strengths. Consider both MALA and MAIMLA applied to a system with quadratic potential, $\beta = 2$, and dimension $N = 300$. For both algorithms we simulate the dynamics (3.1), initialising the system at the limiting distribution of the pure quartic β -ensemble (refer to Figure 3). The step size is chosen according to the linear fit in Figure 15 and we run the systems for an effective time of $T = 10.0$. One experiment considers $M = 50$ independent realisations and another considers $M = 200$.

Every 20 timesteps all M realisations are collected together and the 1-Wasserstein distance between this empirical distribution and the semicircle cdf is calculated (note that the 1-Wasserstein distance can be calculated without the inverse cdf; see, for instance, [94]). These results are plotted in Figure 16. Now, we have chosen the quadratic potential so that we can compare these samples to exact samples drawn from the tridiagonal random matrix model [25]. Indeed, for both choices of M , we draw M independent sets of $N = 300$ eigenvalues from this model, collect all realisations together, and again compute the 1-Wasserstein distance to the semicircle cdf. This is repeated 100 times, with the mean and variance (± 2 standard deviations) superimposed onto the figure. We see that once the Langevin schemes reach equilibrium, they become statistically indistinguishable from the exact samples, indicating that the bias is being removed by the Metropolis step.

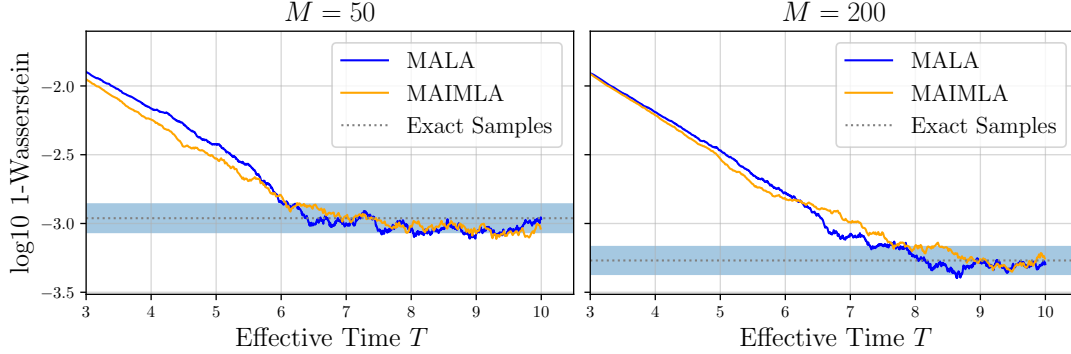


Figure 16: The 1-Wasserstein distance between the semicircle distribution function and samples, collected every 20 steps, from the dynamics (3.1) with a quadratic potential, $\beta = 2$, and dimension $N = 300$. For comparison we also provided the mean (dotted line) with a variance of ± 2 standard deviations (highlighted) of the same distance between M exact samples from the tridiagonal random matrix model [25] over 100 trials.

With the same samples we also investigate the microscopic statistics of the spectrum. In Figure 17 we plot the normalised spacings (2.22) between consecutive eigenvalues for the $M = 200$ cases. The histogram is obtained by considering samples after $T = 8$ and, as before, we only consider the middle 80% of eigenvalues to avoid the chaotic fluctuations around the edge. For these methods the histograms agree almost perfectly with the theoretical distribution - a clear advantage over the unadjusted implicit method discussed in Section 4.3. For fluctuations around the edge we consider a slightly different experiment. Indeed, these statistics are known to relax much more slowly than the macroscopic or microscopic statistics; see, for instance, [31, 54]. Furthermore, it is more important to avoid correlation between the samples. Thus we reduce the dimension to $N = 99$, choose $\beta = 1$, and simulate the same dynamics with MALA for an effective time of $T = 500.0$. We collect samples past a burn-in of $T = 100.0$ every 200 steps and use the scaling

$$x = \frac{\sqrt{N/2}\lambda_{\max} - \mu_N}{\tau_N} \quad (4.7)$$

with constants

$$\mu_N = \sqrt{2 \left(N + \frac{1}{2}\right) - \frac{1}{5} \left(N + \frac{1}{2}\right)^{-1/3}}, \quad \tau_N = \frac{1}{\sqrt{2}(N+1)^{1/6}}$$

instead of (2.24) for an improved $\mathcal{O}(N^{-2/3})$ convergence to the Tracy-Widom distribution [54]. A histogram of the empirical maximal eigenvalues under the transformation (4.7) is given in Figure 18. It exhibits close agreement with the Tracy-Widom distribution $F_1(x)$ superimposed in red.

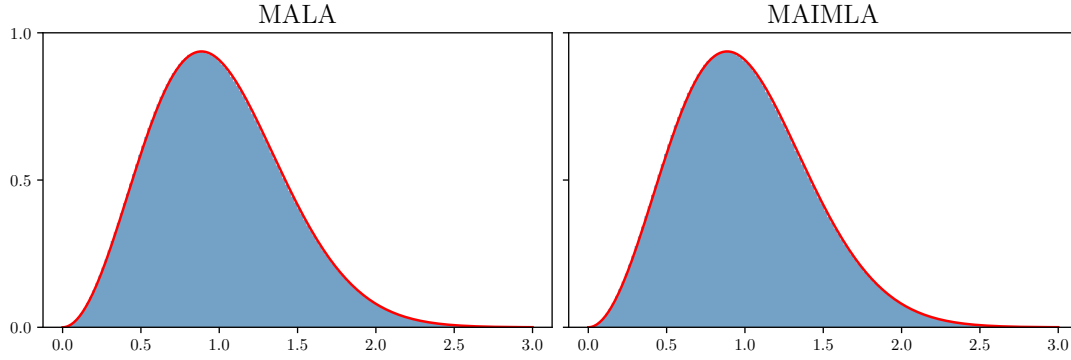


Figure 17: Normalised spacings (2.22) between eigenvalues from the same experiment as in Figure 16. Both plots correspond to the $M = 200$ case and samples are only collected from the middle 80% of eigenvalues after the effective time $T = 8.0$. Superimposed in red is Wigner's surmise for $\beta = 2$ and histograms are plotted without edges and 500 bins.

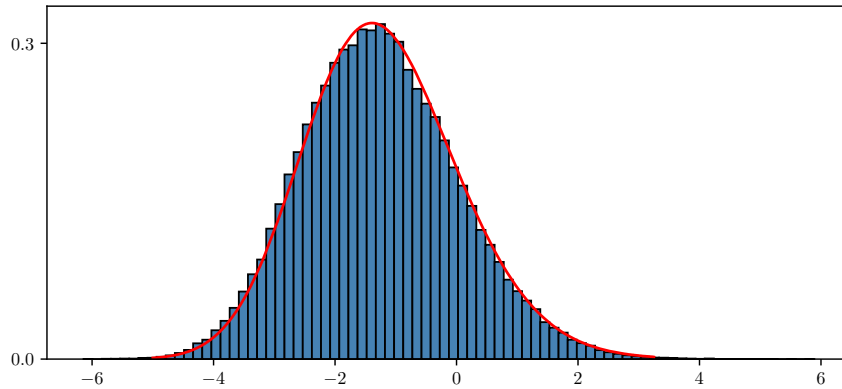


Figure 18: The maximal eigenvalue, scaled under (4.7), of samples drawn every 200 steps from a long run of MALA to a system with $N = 99$, $M = 50$, and $\beta = 1$. The histogram uses 80 bins and in red is the Tracy-Widom distribution for $\beta = 1$, obtained through the TRACYWIDOM Python package.

4.5 On the Ginibre and Wishart-Laguerre Ensembles

Let us now consider the Ginibre and Wishart-Laguerre ensembles as described in Section 2.6. It is in this chapter that we would like to emphasise that the versatility of DBM as a sampling scheme is perhaps its greatest strength. Indeed, it can sample from the β -ensembles in full generality while also being easily extendable to the Ginibre ensembles (and, more generally, higher-dimensional log-gases). For instance, to sample from the Wishart-Laguerre (β -Laguerre) ensembles, all we need to do is modify the potential so that

$$V(x) = \frac{x}{c} - \left(\frac{1}{c} - 1 \right) \ln(x), \quad 0 < c \leq 1.$$

Similarly, to sample from the complex Ginibre ensemble we simply treat the eigenvalues as two dimensional and modify the potential and Coulomb interactions accordingly. To be clear, we treat $\lambda = x + iy$ and modify the potential so that $V(\lambda) = |\lambda|^2 = x^2 + y^2$ and

$$\log |\lambda_k - \lambda_j| = \frac{1}{2} \log [(x_k - x_j)^2 + (y_k - y_j)^2],$$

The gradient correspondingly becomes

$$\nabla_{\lambda_k} \log |\lambda_k - \lambda_j| = \frac{\lambda_k - \lambda_j}{|\lambda_k - \lambda_j|^2} = \frac{1}{(\lambda_k - \lambda_j)},$$

where $\bar{\lambda} := x - iy$. From a log-gas perspective this is now a true Coulomb kernel.

We first run simulations targeting the Wishart-Laguerre ensembles with $c = 1/8$ and $c = 1/32$. Our chosen scheme is MALA, applied to a system of dimension $N = 100$ initialised uniformly on $[(1 - \sqrt{c})^2, (1 + \sqrt{c})^2]$. The empirical density of samples collected every ten steps in the second half of a trajectory with effective time $T = 10.0$ is plotted against the limiting distribution in Figure 19. In both instances there is strong agreement between the samples and the limiting law.

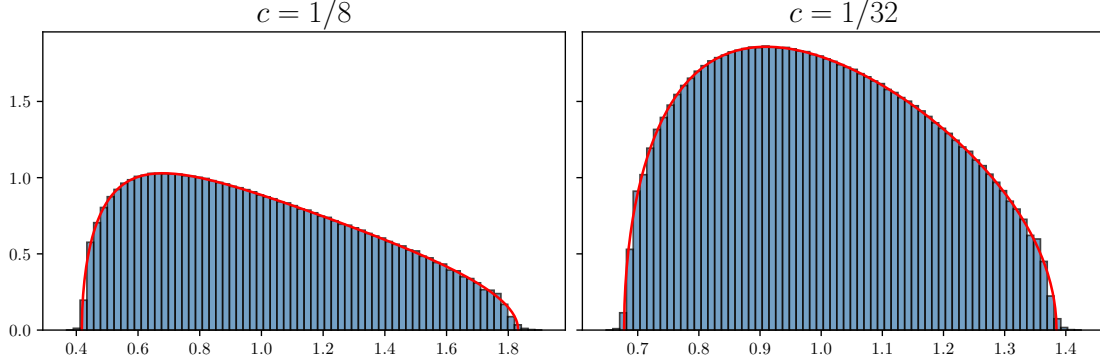


Figure 19: Samples from the Wishart-Laguerre ensembles with $c = 1/8$ and $c = 1/32$, obtained by simulating the dynamics (3.1) with MALA. In both cases $N = 100$, there are $M = 200$ independent trials, the system is initialised uniformly on $[(1 - \sqrt{c})^2, (1 + \sqrt{c})^2]$ and ran until $T = 10.0$. The histograms have 80 bins and are formed from snapshots taken every 10 steps after $T = 3.0$.

A more interesting example is the complex Ginibre ensemble. For one, the rate of convergence to the limiting density is far slower than for the β -ensembles, being of $\mathcal{O}(N^{-1/2})$ rather than $\mathcal{O}(N^{-1})$ [15]. This is evidenced, for example, by the plots of the authors of the one-step HMC algorithm [18] that see histograms far from the limiting law (these authors have $N = 8$ and $N = 50$). Moreover, it is not clear to us if the scaling for MALA changes in two dimensions, as the problem of two eigenvalues crossing completely disappears! To this end we consider a large system of dimension $N = 5000$, initialised from equilibrium, with just $M = 5$ independent realisations. We set $\Delta t = 1/(2\sqrt{N})$ and run the system for an effective time of $T = 3.0$. Samples are collected every three steps in the second half of the trajectory. Figure 20 provides the empirical radial density of these samples against the limiting distribution $f(r) = 2r$. The results show a much closer agreement to the limiting density than those in [18]. Notably, the $\mathcal{O}(N^{-1/2})$ step size produces an acceptance rate of over 50%! This is a drastic improvement over the one-dimensional case and provides further evidence of the importance of eigenvalues (not) crossing. This finding warrants further investigation into how each sampling scheme behaves in higher dimensions. Indeed, if a scheme is of suitable precision and efficiency it could be used for insight into open problems such as the Abrikosov lattice conjecture (see e.g. [78, 79]), a problem that concerns the behaviour of the Coulomb gas as $\beta \rightarrow \infty$.

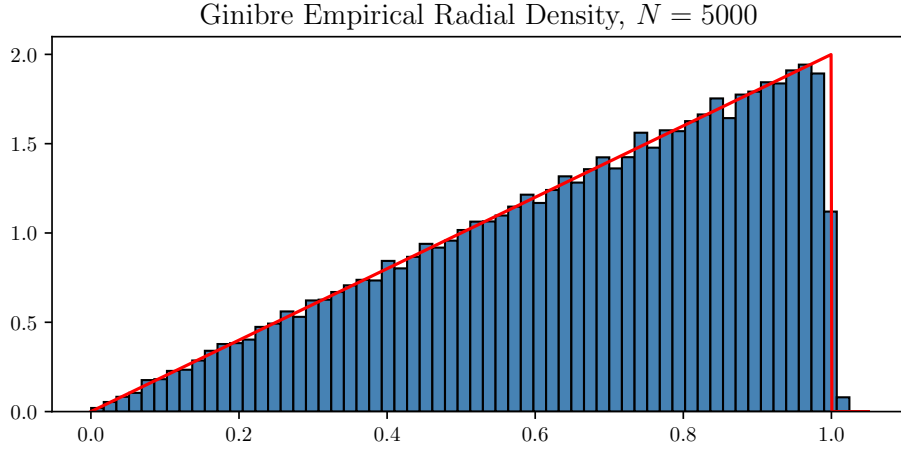


Figure 20: A histogram of the radial density $r = |\lambda|$ of samples from MALA targeting the complex Ginibre ensemble of dimension $N = 5000$. There are $M = 5$ independent trials and snapshots are taken every three steps in the second half of a trajectory with effective time $T = 3.0$. The histogram is formed with 80 bins.

4.6 Fast Multipole Methods and Stochastic Trace Estimation

To conclude our numerical experiments we test the scaling of the fast multipole method (FMM, Section 3.5) and the stochastic Lanczos quadrature algorithm (SLQ, Section 3.6). Recall that the former produces an approximation for the Coulomb interaction

$$\frac{1}{N} \sum_{j \neq k} \frac{1}{\lambda_k - \lambda_j}$$

for all particles λ_k with $\mathcal{O}(N)$ complexity [42] instead of the direct $\mathcal{O}(N^2)$ evaluation. The latter is an algorithm that approximates the log-determinant of a matrix, namely for the difference

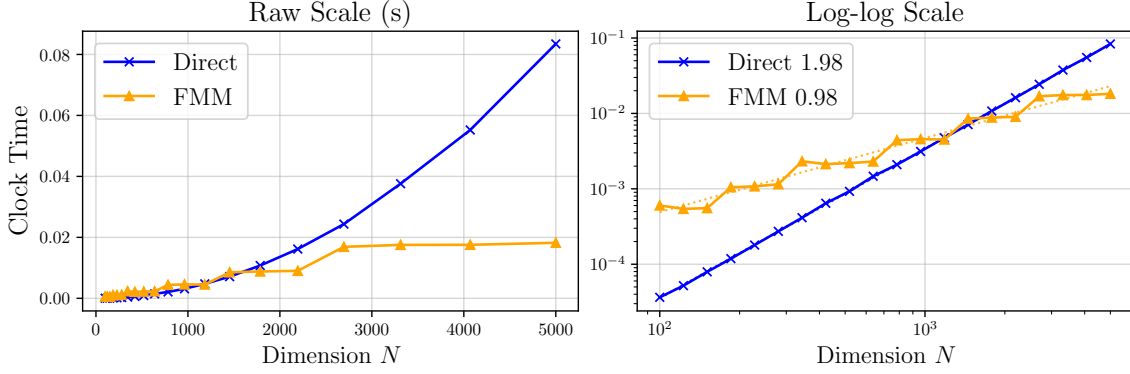
$$\log(|\det(I_N + \Delta t \nabla^2 H_N(x))|) - \log(|\det(I_N + \Delta t \nabla^2 H_N(y))|)$$

that appears in MAILA’s transition kernel (3.21). It too is an approximation but is asymptotically of $\mathcal{O}(N^2)$ complexity [93], an improvement over the Cholesky factorisation method [91] which is of order $\mathcal{O}(N^3)$. Both FMM and SLQ have a significant amount of computational overhead attached to them, and a large part of the motivation for these experiments is to find the threshold for when this overhead is outweighed by the improved complexity. For each algorithm we perform the following procedure.

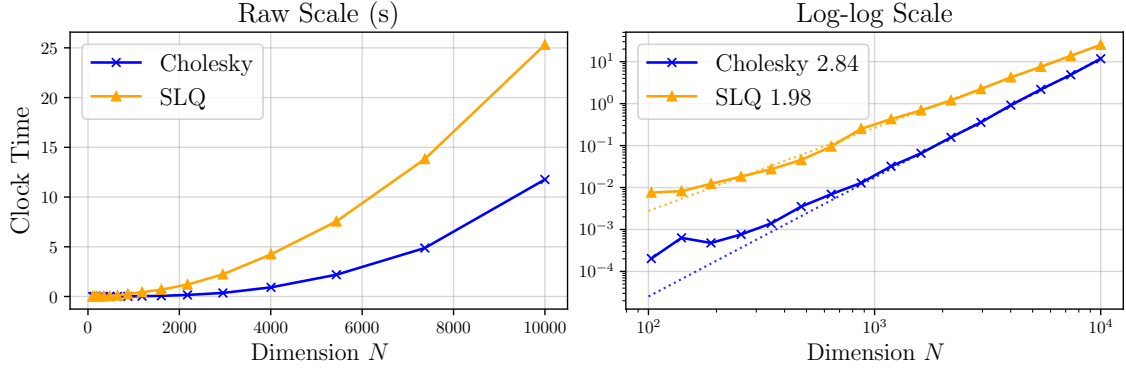
1. Choose a range of dimensions N .
2. Record the clock time it takes for each method (e.g. direct vs FMM, Cholesky vs SLQ) to perform its respective calculation (Coulomb and log-determinant).
3. Average these scores across M independent trials.

For the Coulomb interaction comparison we choose N log-spaced over the interval $[100, 5000]$ and average the clock times over $M = 1000$ trials. We run FMM with the parameters $p = 10$ and $s = 32$. Recall that p indicates the number of terms taken in various truncated infinite sums and influences the precision of the algorithm. Similarly, s indicates the number of particles in each of the tree’s leaves, indirectly influencing the precision through the number of particles treated as ‘near’ and thus calculated exactly. For the determinant calculation, we choose N log-spaced over the interval $[100, 10000]$ and choose a smaller number of $M = 5$ trials. We set the parameters $m = 30$ and $K = 20$, where K is the number of Rademacher vectors and m is the size of the quadrature rule. These are in line with parameters chosen by the original authors for systems of similar size [93].

The results of this experiment can be seen in Figure 21. Notice that the empirical scaling matches the theoretical scaling in all four cases. These results suggest that FMM is only of utility for $N \gtrsim 2000$, but sees significant improvement thereafter. On the other hand, the Cholesky factorisation approach for the log-determinant appears to outperform SLQ even for $N = 10000$. Therefore we suggest that SLQ does not have much practical use for this problem because for systems so large one would likely use MALA or MAIMLA instead, as these schemes have comparable scaling (as seen in Section 3.3) and a cheaper per-step cost.



(a) Direct calculation versus the fast multipole method (FMM) with parameters $p = 10$ and $s = 32$. Evaluated for N log-spaced over the interval $[100, 5000]$ with each time averaged over $M = 1000$ independent trials. The log-slope is indicated in the legend.



(b) Cholesky factorisation versus the stochastic Lanczos quadrature (SLQ) with parameters $m = 30$ and $K = 20$. Evaluated for N log-spaced over the interval $[100, 10000]$ with times averaged over $M = 5$ independent trials. The log-slope is indicated in the legend.

Figure 21: Clock time for different approaches to the Coulomb interaction (top panel) and the log-determinant (bottom-panel).

To finish, we run a small experiment that tests the precision of FMM. Consider leaf sizes $s \in \{32, 64\}$, multipole orders $p \in \{3, 5, 10, 20\}$, and $N \in \{2000, 5000, 10000, 20000\}$. For each dimension N we compute eigenvalues from the GUE (through the tridiagonal random matrix model [25]) and evaluate the exact Coulomb interaction. Then we calculate the maximum absolute error over 100 trials between this exact evaluation and the FMM approximation with parameters (s, p) . These results are given in Table 3. Observe that the error seems to depend mainly on p and has very mild dependence on N , being as small as 10^{-9} with $p = 20$. Indeed, we suggest that the approximation error is negligible and certainly not a limiting factor of the algorithm.

s	N	$p = 3$	$p = 5$	$p = 10$	$p = 20$
32	2000	$9.150\,32 \times 10^{-3}$	$1.000\,90 \times 10^{-3}$	8.5069×10^{-6}	1.1×10^{-9}
	5000	$1.067\,51 \times 10^{-2}$	$1.312\,39 \times 10^{-3}$	$1.201\,58 \times 10^{-5}$	1.8×10^{-9}
	10000	$1.157\,26 \times 10^{-2}$	$1.371\,24 \times 10^{-3}$	$1.243\,17 \times 10^{-5}$	1.8×10^{-9}
	20000	$1.285\,88 \times 10^{-2}$	$1.492\,59 \times 10^{-3}$	$1.398\,42 \times 10^{-5}$	2.2×10^{-9}
64	2000	$8.367\,08 \times 10^{-3}$	$8.989\,43 \times 10^{-4}$	7.2091×10^{-6}	9.0×10^{-10}
	5000	$1.004\,73 \times 10^{-2}$	$1.215\,78 \times 10^{-3}$	$1.115\,03 \times 10^{-5}$	1.7×10^{-9}
	10000	$1.085\,87 \times 10^{-2}$	$1.309\,07 \times 10^{-3}$	$1.198\,09 \times 10^{-5}$	1.8×10^{-9}
	20000	$1.204\,02 \times 10^{-2}$	$1.421\,46 \times 10^{-3}$	$1.328\,72 \times 10^{-5}$	2.1×10^{-9}

Table 3: The maximum absolute error between the exact Coulomb interaction and the FMM approximation, over 100 independent sets of eigenvalues from the GUE, for varying multipole order p , leaf size s , and system dimension N .

5 Conclusion

This dissertation set out with two primary objectives: to provide an accessible introduction into random matrix theory, and to investigate the utility of Dyson Brownian motion as a sampling scheme. We conclude with a discussion of key ideas from both themes.

Let us first comment on random matrices. We feel that they are very interesting objects to study! The fact that one can start with independent matrix entries yet observe strong correlation between eigenvalue statistics is incredible. This notion is only made more fascinating by the idea of universality - that different scales of eigenvalue statistics depend only on some aspects of the random matrix construction. For instance, we have seen that the bulk (macroscopic) statistics are controlled by the potential V , whereas the local (microscopic) statistics are controlled by β . Often we have understood V and β from a statistical mechanics perspective through the energy

$$H_N(x) = \frac{1}{2} \sum_{j=1}^N V(x_j) - \frac{1}{N} \sum_{j < k} \log(x_k - x_j)$$

and it is in this framework that we understand Dyson Brownian motion, mainly for the purpose of sampling from random matrix ensembles through the simulation of Langevin dynamics. Its versatility makes it an attractive approach; indeed, its scope of random matrix ensembles is far greater than alternative schemes such as the tridiagonal random matrix model or determinantal point process sampling. However, the logarithmic repulsion produces a large source of computational difficulty. An interesting result we have found is that this difficulty materialises in different ways for different schemes: explicit schemes struggle with the problem of two trajectories crossing, whereas implicit schemes face a more subtle challenge in the Hessian conditioning. On a practical note, we propose that the scheme used to simulate the dynamics should be chosen with consideration to the statistics of interest. We have seen that the implicit Langevin algorithm has a desirable complexity of $\mathcal{O}(N^2)$ and recovers the macroscopic statistics well, but for practical sampling of more delicate behaviour such as the local spacings or edge fluctuations, one should use an unbiased algorithm. In this case the classical option of MALA seems to perform well, particularly because it can be paired with a fast multipole method to achieve the sub-cubic complexity $\mathcal{O}(N^{5/3} \cdot N) = \mathcal{O}(N^{8/3})$. Both of these algorithms improve on the tamed Euler scheme [57] and, up to interpretation of its equivalence, the one-step Hamiltonian Monte Carlo scheme [18]. Moreover, they are more efficient than the determinantal point process sampler [69], far more applicable than the tridiagonal model [25], and comparable in efficiency to the Gibbs tridiagonal sampler [36] with the added benefit of extending to higher dimensional systems.

There are several interesting avenues for future work on the topic of this dissertation. For example, there are particular instances of missing analysis that would aid understanding of implicit methods for the problem, namely their convergence on restricted domains such as the Weyl chamber and the apparent insensitivity of the macroscopic eigenvalue statistics to bias. More generally, we would like to understand if the implicit midpoint method has any advantageous properties for the sampling problem, as it can benefit from strengths native to both explicit and implicit methods. It is also natural to consider how the derived step size scaling generalises to different ensembles. We have mainly performed analysis on the β -ensembles for only select potentials, and we wonder if there is any general dependence on β or V . Perhaps more importantly, we suggest that systems living in dimensions $d \geq 2$ are of interest because there the major problem of trajectories crossing vanishes! Indeed, for $d = 2$ the scaling $\Delta t = \mathcal{O}(N^{-1/2})$ seemed to suffice for effective sampling. An efficient algorithm here could conceivably be applied to more general problems for log-gases and interacting particle systems.

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